Hybrid Domain Joint Low-Rank and Sparse-Constrained Framework for Simultaneous RFI Separation and SAR Imaging

Jun'ao Li, *Member, IEEE*, Zhongyu Li, *Senior Member, IEEE*, Jing Yang, *Member, IEEE*, Qing Yang, *Member, IEEE*, and Junjie Wu, *Senior Member, IEEE*

***Abstract*—The inherent contradiction between limited spectrum resources and ever-growing spectrum demand makes it difficult for synthetic aperture radar (SAR) to avoid in-band radio frequency interference (RFI) during observation, leading to significant degradation in imaging quality. In this context, RFI suppression technology has become crucial for obtaining high-fidelity SAR images. Existing RFI suppression methods are primarily divided into preprocessing and postprocessing categories, both of which have inherent limitations. Preprocessing methods often struggle to achieve sufficient suppression due to the high overlap between target echoes and RFI in the time, frequency, time–frequency, and other domains. Postprocessing methods, which operate directly on the imaging results, tend to cause loss of image details and weak targets. Motivated by these problems, this article proposes a hybrid domain joint low-rank and sparse-constrained framework for simultaneous RFI separation and SAR imaging. The innovation lies in leveraging the low-rank characteristic of RFI in the echo domain as well as the low-rank and sparse representation of scene targets, while coupling the echo and image domains through the SAR imaging operator to establish a unified inversion model with relational constraints. To solve this model, a deep learning network named SSI-net is constructed by integrating the alternating direction method of multipliers (ADMMs) with deep unfolding techniques. Besides, for scenarios with limited or no training samples, a gradient descent-based optimization algorithm is also proposed. Simulations and experiments with measured data demonstrate the superior performance of the proposed framework.**

***Index Terms*—Gradient descent, hybrid domain, joint low rank and sparse, radio frequency interference (RFI) separation and synthetic aperture radar (SAR) imaging, SSI-net.**

I. Introduction

SYNTHETIC aperture radar (SAR) is capable of all-day and all-weather high-resolution imaging, unaffected by adverse weather conditions such as clouds, rain, and fog. It has been widely applied in both civilian and military domains, including Earth mapping, disaster monitoring and assessment, and battlefield reconnaissance and surveillance [1], [2], [3]. However, the severe scarcity of spectral resources, coupled with growing human demand, has led to increasingly intense competition. Against this backdrop, the electromagnetic environment has become more complex, making SAR systems highly susceptible to radio frequency interference (RFI) from various sources, such as communication and radar devices operating in the same frequency band, which severely degrades image quality [4], [5], [6], [7].

In March 2023, NASA released RFI monitoring results from the SMAP mission [8], which is shown in Fig. 1. The color indicates RFI persistence. The data reveal severe RFI pollution in Europe, the Middle East, and East Asia. In extreme cases, interference can completely mask targets in the scene, making it impossible to extract information about regions of interest. In response, researchers have conducted extensive studies on RFI mitigation [9], [10], [11], [12], which can be broadly categorized into two groups: preprocessing- and postprocessing-based interference suppression methods.

Preprocessing-based interference suppression methods operate directly on the echoes. These methods first perform RFI suppression on the collected data to separate useful target echoes from the beam-illuminated scene, followed by the application of imaging algorithms to produce high-quality SAR images. After years of sustained and in-depth research by scholars, preprocessing methods have been further refined into categories such as parametric modeling [13], filtering [14], [15], and feature-decomposition-based techniques [16], [17].

Parametric-model-based methods operate on the core principle of precisely modeling the RFI signal. These techniques estimate relevant parameters, such as amplitude and frequency, from echoes [18], [19], [20], and subsequently subtract the estimated RFI component to achieve interference suppression. Consequently, the accuracy of both RFI modeling and parameter estimation is directly linked to the final suppression performance. However, as the electromagnetic environment grows increasingly complex, characterizing RFI using deterministic parametric models has become significantly more challenging. Model mismatches under these conditions would lead to severe performance degradation in RFI suppression.

Filter-based methods suppress RFI by transforming the echoes into a domain, such as the frequency domain (FD) [21]

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Jun'ao Li and Jing Yang are with the School of Electrical and Information Engineering, Zhengzhou University, Zhengzhou 450001, China (e-mail: lijunao@outlook.com).

Zhongyu Li, Qing Yang, and Junjie Wu are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: zhongyu_li@hotmail.com; yangqinguestc@outlook.com).

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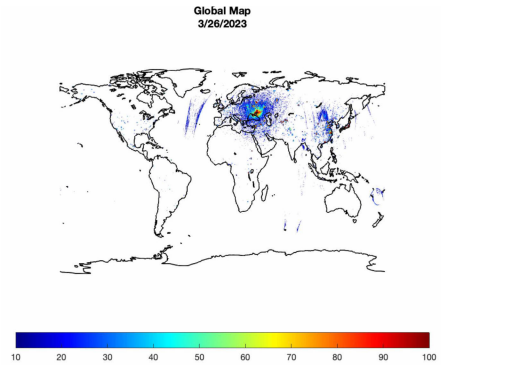

Fig. 1. Distribution of RFI in the worldwide (NASA).

or time–frequency domain (TFD) [22], where the distributed interference energy becomes concentrated. A matched filter is then constructed in the corresponding domain to realize RFI suppression. However, this filtering process often inadvertently removes a portion of target echoes, leading to elevated sidelobe levels and resolution reduction in the final SAR image.

Feature-decomposition-based methods lie in leveraging the distinct characteristics of target echoes and RFI signals, such as low-rank and sparse properties [23], [24], [25], [26], to establish and solve a separation model for extracting useful target echoes. In these approaches, RFI is typically modeled as low rank or sparse, based on its narrowband and spectrally sparse nature [27], [28], [29], [30], while the useful target echoes are modeled as sparse in the range dimension under the assumption of few targets. However, the applicability of this target sparsity assumption in SAR systems is conditional. It generally holds for maritime vessels, ground-moving targets, or isolated strong scatterers [31]. In practice, the beam footprint of SAR often encompasses distributed background targets like grasslands, hills, and buildings, which produce continuously distributed echoes in the range dimension, with similarly continuous energy trajectories in the azimuth dimension. Consequently, it is difficult to characterize SAR target echoes using simple sparsity, making low-rank and sparsity-based methods such as robust principal component analysis (RPCA) less effective for RFI suppression than desired. Besides, driven by advances in deep learning, some researchers have sought to leverage neural network to extract distinctive features of RFI and target echoes for effective separation. For instance, Fan et al. [32] designed an end-to-end interference mitigation network (IMN) that employs a data-driven approach to separate target echoes in the TFD. Lou et al. [33], proposed a feature-enhanced low-rank and sparse decomposition network (FELS-net), which learns low-rank transformation of target echoes and sparse transformation of RFI to achieve interference suppression. However, these deep-learning-based methods require the construction of datasets for training. When there is a distribution shift between training data and the actual operational environment, the model would suffer from significant performance degradation, leading to severely compromised image quality after interference suppression.

Postprocessing RFI suppression methods operate on the complex-valued SAR image that has been contaminated by RFI. These techniques aim to suppress RFI by exploiting characteristic differences between the interference and SAR targets directly in the image domain. For example, a principal-component-analysis-based approach was proposed in [34], while another method in [35] developed a 2-D spectral analysis (SPECAN) analysis (2DSA) filter to excise RFI with chirp-like characteristics through spectral-domain filtering. However, in practical scenarios where RFI types are highly diverse, these methods struggle to effectively suppress other interference categories, exhibiting significant adaptability issues. To address this, a generic subspace model was proposed in [36] to characterize various RFI types. This approach designed a block subspace filter (BSF) by leveraging the low-dimensional structure of the RFI subspace for suppression. However, it is assumed that the observed scene contains no strong point scatterers. Violation of this assumption leads to severe attenuation of these scatterers' intensities. A lambda-1 detector was constructed in [37] that can adaptively identify and filter out eigenvalues in the complex image caused by RFI, thereby improving image quality. Nevertheless, this technique requires strong scatterers to be removed in advance, as their presence will bias the detector's threshold estimation, often resulting in the loss of image details. Alternatively, the method in [38] utilizes multiple SAR images of the same scene acquired at different times to form an interferometric SAR pair, attempting to reconstruct a clean SAR image from the interferogram. A fundamental constraint of this approach is its inability to perform reconstruction when the interferometric conditions are poor or when only a single SAR image is available.

Inspired by these challenges, this article proposes a framework for simultaneous RFI separation and SAR imaging. The proposed framework directly utilizes the acquired RFI-contaminated echo data to achieve high-precision imaging and RFI separation without additional preprocessing or postprocessing. The implementation and main contributions of the framework are as follows.

1) A bridge between the echo and image characteristic constraints is established through the SAR imaging operator. This enables the construction of a hybrid domain joint low-rank and sparse-constrained model for simultaneous RFI separation and SAR imaging.
2) The closed-form solutions for all subproblems within this model using the alternating direction method of multipliers (ADMMs) are derived. Based on this solving process, a deep unfolding network named SSI-net is constructed, which effectively mitigates performance degradation caused by inappropriate manual hyperparameter settings while accelerating the convergence speed of the solution process.
3) To address scenarios with limited or no training samples, a gradient-descent-based optimization method for simultaneous RFI separation and SAR imaging is also proposed. Although this approach sacrifices some computational efficiency, it achieves performance nearly consistent with that of SSI-net.

The remainder of this article is organized as follows. Section II presents the signal model and provides characteristic analysis of both RFI and target echoes. Section III

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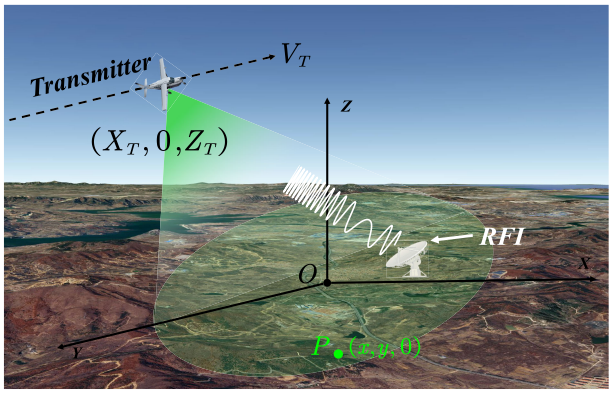

Fig. 2. SAR geometric configuration.

elaborates on the implementation workflow of the proposed framework for simultaneous RFI separation and SAR imaging. Section IV offers a comprehensive performance evaluation through numerical simulations and airborne flight experiments, presenting quantitative comparisons with existing methods in terms of interference suppression capability, which collectively demonstrate the superior performance of the proposed framework. Finally, Section V gives the conclusion.

II. Signal Model and Characteristic Analysis

This section establishes the signal model for SAR echoes, narrowband, and wideband RFI, analyzes their respective characteristics, and presents traditional RFI mitigation models. It lays the necessary foundation for introducing the subsequent simultaneous RFI separation and SAR imaging framework.

A. Signal Model

The operational geometry of the SAR system is illustrated in Fig. 2. At azimuth zero moment, the beam center O is set as the origin. The Y-axis is aligned parallel to the platform's flight direction, the Z-axis is perpendicular to the ground plane, and the X-axis is orthogonal to the YOZ plane, thereby establishing a Cartesian coordinate system. The transmitter's coordinates and velocity are denoted as $(X_T, 0, Z_T)$ and V_T, respectively. P denotes a point target within the beam-illuminated scene, with $(x, y, 0)$ representing its corresponding coordinates.

Assuming that the transmitted signal is $s(t)$, then the received baseband echo after demodulation is

$$s(t,\tau) = \sigma\omega_r\left[\tau - \frac{2R(t)}{c}\right]\omega_a\left(\frac{t-t_a}{T_s}\right) \times s\left[\tau - \frac{2R(t)}{c}\right]\exp\left\{j2\pi f_c\left[\tau - \frac{2R(t)}{c}\right]\right\} \quad (1)$$

where $\omega_r[\cdot]$ and $\omega_a(\cdot)$ denote the range and azimuth envelopes, respectively. σ is the amplitude. t and τ represent the slow time and fast time, respectively. t_a and T_s indicate the beam center moment and synthetic aperture time, respectively. f_c and c stand for the signal carrier frequency and speed of light, respectively. The range between the transmitter and target P, denoted as $R(t)$, is given by

$$R(t) = \sqrt{(X_T - x)^2 + (V_T t - y)^2 + Z_T^2}. \quad (2)$$

TABLE I
RFI Simulation Parameters

RFI types	Symbol	Value
$I_{nbi}(t)$	K	3
	$f_{n,k}$	[9.3,9.6,9.9]GHz
	φ_k	[0,2π]
$I^l_{wbi}(t)$	K	2
	$f_{w,k}$	[9.32,9.72]GHz
	γ_k	$[1.5\times10^{13}, 2\times10^{14}]$
	φ_k	[0,2π]
$I^s_{wbi}(t)$	K	2
	$f_{w,k}$	[9.6,9.8]GHz
	α_k	[10,100]
	$f^s_{w,k}$	[0.5,2]MHz
	φ_k	[0,2π]

On the other hand, when the frequency spectrum of RFI overlaps with the SAR operating frequency band, the SAR imaging result will be severely degraded, posing significant challenges for subsequent processing tasks such as detection and recognition. Based on the relative bandwidth of the radiating source, RFI can be categorized into narrowband interference (NBI) and wideband interference (WBI). Specifically, NBI can be modeled as a combination of multiple single-frequency components, which is expressed as

$$I_{nbi}(t) = \sum_{k=1}^{K}\zeta_k(t)\, e^{j(2\pi f_{n,k}t+\varphi_k)} \quad (3)$$

where $\zeta_k(t)$, $f_{n,k}$, and φ_k represent the amplitude, carrier frequency, and initial phase of the kth NBI, respectively. K denotes the total number of interference sources.

And there are two common modulation schemes for WBI, which are linear frequency modulation (LFM) I^l_{wbi} and sinusoidal frequency modulation (SFM) I^s_{wbi}. Their models are defined as

$$I^l_{wbi}(t) = \sum_{k=1}^{K}\zeta_k(t)\, e^{j\pi(2f_{w,k}t+\gamma_k t^2+\varphi_k)} \quad (4)$$

$$I^s_{wbi}(t) = \sum_{k=1}^{K}\zeta_k(t)\, e^{j\left[\alpha_k\sin(2\pi f^s_{w,k}t+\varphi_k)+2\pi f_{w,k}t\right]} \quad (5)$$

where $f_{w,k}$ and γ_k denote the carrier frequency and chirp rate of the kth I^l_{wbi}, while α_k and $f^s_{w,k}$ represent the modulation factor and modulation frequency of the kth I^s_{wbi}, respectively.

B. Echo Characteristic Analysis

Guided by the RFI models defined in (3)–(5) and the parameters listed in Table I, corresponding RFI signals are simulated. The distinct LFM and SFM characteristics of these signals are visualized in the TFD, as shown in Fig. 3.

In the TFD, LFM WBI appears as an inclined straight line whose slope corresponds to its chirp rate. In contrast, SFM wideband interference resembles a sinusoidal curve. Due to its typically slow variation along the slow-time dimension,

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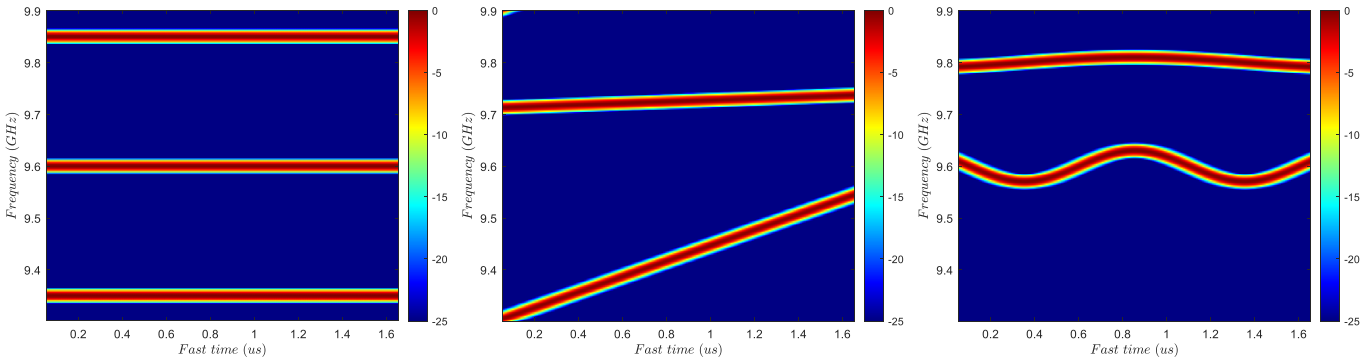

Fig. 3. RFI characteristic in time–frequency domain. (a) NBI. (b) LFM WBI. (c) SFM WBI.

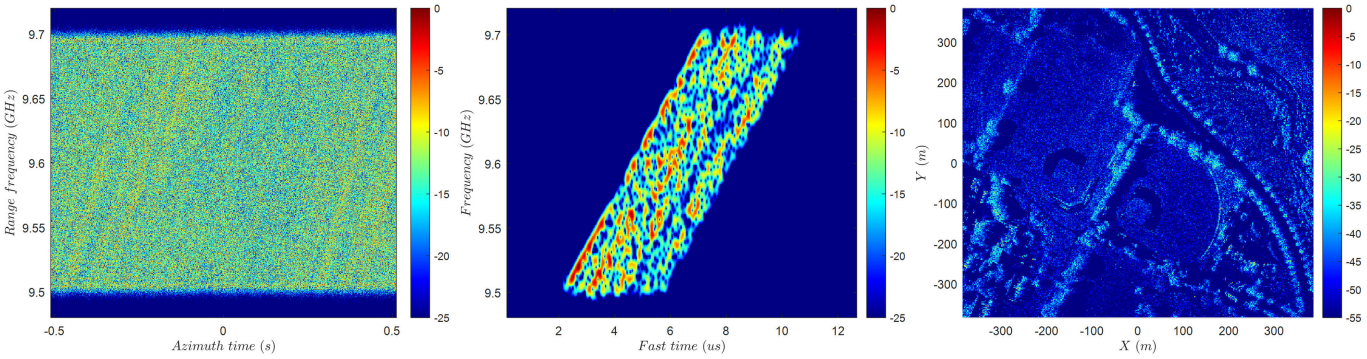

Fig. 4. SAR echo and imaging result. (a) Echo in range–FD. (b) Echo in TFD. (c) Imaging result.

RFI generally exhibits low-rank characteristics, as discussed in prior works [23], [24].

On the other hand, the SAR system processes the acquired echoes to generate high-resolution images of the illuminated scene. For the simulation, a transmitter located at (−5000, 0, 5000) m moves with a velocity of 150 m/s. The transmitted signal is an LFM pulse with a width of 5 μs, a bandwidth of 500 MHz, and a center frequency of 9.6 GHz. Fig. 4 presents the corresponding echo data in both the range–FD and TFD, as well as the final imaging result. The image in Fig. 4(c) is generated using the range–Doppler (RD) algorithm [39], a process that can be formally represented by the following linear operator:

$$\begin{aligned}\mathbf{X} &= \mathcal{H}\{\mathbf{Y}\} \\ &= F_a^{-1}\left\{M\left\{F_a\left\{F_r^{-1}\{\mathbf{Y}\odot\mathbf{H}_r\}\right\}\right\}\odot\mathbf{H}_a\right\}\end{aligned} \tag{6}$$

where $\mathcal{H}\{\cdot\}$ represents the RD imaging operator. $\mathbf{H}_r$ and $\mathbf{H}_a$ denote the range and azimuth matched filters, respectively. $M\{\cdot\}$ is the range cell migration correction operator. $F_a\{\cdot\}$ and $F_a^{-1}\{\cdot\}$ are the azimuth fast Fourier transform (FFT) and inverse FFT, respectively. $F_r\{\cdot\}$ and $F_r^{-1}\{\cdot\}$ represent the range FFT and inverse FFT, respectively. $\odot$ means the Hadamard product. $\mathbf{X}$ and $\mathbf{Y}$ indicate the image result and target echoes in the range–FD, respectively.

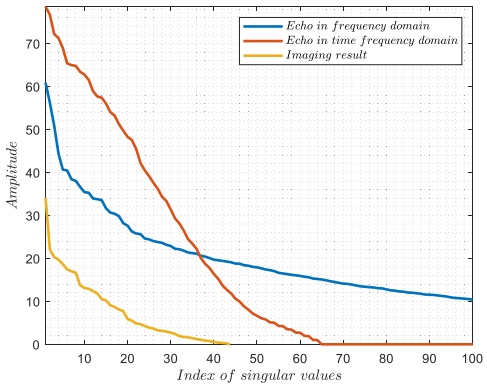

Fig. 5. Singular value curves of echoes and imaging result.

Consequently, the imaging inverse process can be deduced, i.e., the echo acquisition procedure, which is expressed as

$$\begin{aligned}\mathbf{Y} &= \mathcal{G}\{\mathbf{X}\} \\ &= F_r\left\{F_a^{-1}\left\{M^{-1}\left\{F_a\{\mathbf{X}\}\odot\mathbf{H}_a^*\right\}\right\}\right\}\odot\mathbf{H}_r^*\end{aligned} \tag{7}$$

where $\mathbf{H}_r^*$ and $\mathbf{H}_a^*$ represent the conjugate forms of range and azimuth matched filters, respectively. $M^{-1}\{\cdot\}$ represents the inverse process of range migration correction operator.

As shown in Fig. 4(a) and (b), the target echoes occupy the majority of the corresponding domains in both range–frequency and time–frequency representations,

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exhibiting no discernible sparsity. In contrast to RFI, the target echoes also deviate from a low-rank structure. This is evidenced by the singular value curves presented in Fig. 5, which exhibit a gradual decay rather than a sharp drop, indicating a less compact energy distribution.

In the presence of RFI and environmental noise, the signal $\mathbf{Y}$ received by the SAR system is expressed as

$$\mathbf{Y} = \mathbf{E} + \mathbf{I} + \mathbf{N} \tag{8}$$

where $\mathbf{E}$, $\mathbf{I}$, and $\mathbf{N}$ represent the target echoes, RFI, and noise components, respectively. Given that the target echoes exhibit sparsity and the RFI exhibits a low-rank structure, a joint suppression model can be formulated as

$$\begin{aligned} &\min \ \ \|\mathbf{E}\|_1 + \kappa \|\mathbf{I}\|_* \\ &\text{s.t.} \ \ \|\mathbf{Y} - \mathbf{E} - \mathbf{I}\|_F^2 \leqslant \varepsilon \end{aligned} \tag{9}$$

where $\|\cdot\|_1$, $\|\cdot\|_*$, and $\|\cdot\|_F$ denote the l_1, nuclear, and Frobenius norms, respectively. κ indicates regularization parameter, and ε means error tolerance.

The optimization model in (9) can be solved using the augmented Lagrangian method. Nevertheless, the sparse target prior assumption inherent to this model is often insufficient for effective RFI separation in SAR systems. Consequently, the residual interference leads to severe degradation of the imaging quality, which adversely affects subsequent tasks such as image interpretation.

III. Simultaneous Rfi Separation and SAR Imaging Framework

This section presents a framework for simultaneous RFI separation and SAR imaging. The proposed framework begins by establishing a joint RFI separation and imaging model, which leverages the low-rank property of RFI in the echo domain and the joint low-rank and sparse characteristics of the imaging result, coupled through the SAR imaging operator. Subsequently, the model is solved by ADMM, leading to the construction of SSI-net. To address scenarios with limited or no training samples, a gradient-descent-based optimization method is also proposed. Both solutions demonstrate effective RFI separation performance and can also achieve high-precision SAR imaging.

A. Simultaneous Rfi Separation and SAR Imaging: Modeling and Solution

Neither the SAR echo nor the imaging result exhibits a purely low-rank or sparse characteristic. As demonstrated in [40], the imaging scene can be precisely represented by a highly correlated low-rank matrix $\mathbf{L}$ and a sparse matrix $\mathbf{S}$ composed of strong scatterers, which is formulated as

$$\mathbf{X} = \mathbf{L} + \mathbf{S}. \tag{10}$$

Combined with the low-rank characteristic of RFI, the hybrid-domain-constrained model can be established by using RD imaging operator, which is expressed as

$$\begin{aligned} &arg \min_{\mathbf{L},\mathbf{S},\mathbf{I}} rank(\mathbf{L}) + \gamma_1 \|\mathbf{S}\|_0 + \gamma_2 rank(\mathbf{I}) \\ &\text{s.t.} \quad \mathbf{Y} = \mathcal{G}\{\mathbf{L} + \mathbf{S}\} + \mathbf{I} \end{aligned} \tag{11}$$

where γ_1 and γ_2 indicate regularization parameters. Nevertheless, the optimization model in (11) is a specific NP-hard problem. To enable simultaneous RFI separation and SAR imaging, the l_0-norm and the rank constraint are relaxed to their standard convex surrogates: the l_1-norm and the nuclear norm, respectively, which are presented as

$$\begin{aligned} &arg \min_{\mathbf{L},\mathbf{S},\mathbf{I}} \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{I}\|_* \\ &\text{s.t.} \quad \mathbf{Y} = \mathcal{G}\{\mathbf{L} + \mathbf{S}\} + \mathbf{I}. \end{aligned} \tag{12}$$

To solve the equivalent model in (12), a Lagrange multiplier matrix $\mathbf{U}$ and a penalty parameter ρ are introduced, and then, the unconstrained augmented Lagrangian function is denoted as

$$\begin{aligned} \mathcal{L}(\mathbf{L},\mathbf{S},\mathbf{I},\mathbf{U},\rho) = &\|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{I}\|_* \\ &+ < \mathbf{U}, \mathbf{Y} - \mathcal{G}\{\mathbf{L} + \mathbf{S}\} - \mathbf{I} > \\ &+ \frac{\rho}{2} \|\mathbf{Y} - \mathcal{G}\{\mathbf{L} + \mathbf{S}\} - \mathbf{I}\|_F^2 \end{aligned} \tag{13}$$

where $\langle \cdot, \cdot \rangle$ denotes the matrix inner product.

ADMM is an iterative algorithm well-suited for separable convex optimization problems, especially those with multiple nonsmooth terms. Its core principle is to decompose the original problem into a sequence of simpler subproblems, which are then solved by alternately updating the primal variables ($\mathbf{L}$, $\mathbf{S}$, and $\mathbf{I}$) and the dual variable $\mathbf{U}$. The detailed iterative procedure is as follows.

1) Update the Low-Rank Component $\mathbf{L}$*:* By fixing the variables $\mathbf{S}$, $\mathbf{I}$, and $\mathbf{U}$, the minimization subproblem with respect to $\mathbf{L}$ can be obtained, which is formulated as

$$\mathbf{L}^{k+1} = \underset{\mathbf{L}}{arg\min} \ \|\mathbf{L}\|_* + \frac{\rho}{2} \left\| \mathbf{Y} - \mathcal{G}\left\{\mathbf{L} + \mathbf{S}^k\right\} - \mathbf{I}^k - \frac{\mathbf{U}^k}{\rho} \right\|_F^2 \tag{14}$$

where $(\cdot)^k$ denotes the value for the kth iteration. Different from the traditional nuclear norm model, (14) involves the imaging operator $\mathcal{G}\{\cdot\}$, making it extremely difficult to solve $\mathbf{L}$. To address this issue, the concept of linearized ADMM is adopted [41], [42]. By performing a first-order Taylor expansion of the quadratic penalty term and adding a proximal term, the specific constraint of $\mathcal{G}\{\cdot\}$ can be decoupled, which is expressed as

$$\begin{aligned} \mathbf{L}^{k+1} = \underset{\mathbf{L}}{arg\min} \ &\|\mathbf{L}\|_* + \underbrace{\frac{\kappa_l}{2} \left\|\mathbf{L} - \mathbf{L}^k\right\|_F^2}_{Proximal\ term} \\ &\underbrace{+ \rho < \mathcal{H}\left\{\mathcal{G}\left\{\mathbf{L}^k + \mathbf{S}^k\right\} + \mathbf{I}^k - \mathbf{Y} + \frac{\mathbf{U}^k}{\rho}\right\}, \mathbf{L} - \mathbf{L}^k >}_{The\ first-order\ Taylor\ expansion} \end{aligned} \tag{15}$$

where κ_l represents the linearization parameter. $\mathcal{G}\{\cdot\}$ and $\mathcal{H}\{\cdot\}$ are the adjoint operators of each other, corresponding to the echo generation process and the imaging process, respectively. The specific expressions are shown in (6) and (7). Then, the subproblem can be further simplified as (16), which is shown at the bottom of the next page.

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The closed-form solution for this subproblem can be obtained using the singular value thresholding (SVT) algorithm, which is derived as

$$\mathbf{L}^{k+1} = \mathcal{D}_{\frac{1}{\kappa_l}}\left(\mathbf{L}^k - \frac{\rho}{\kappa_l}\mathcal{H}\left\{\mathcal{G}\left\{\mathbf{L}^k + \mathbf{S}^k\right\} + \mathbf{I}^k - \mathbf{Y} + \frac{\mathbf{U}^k}{\rho}\right\}\right) \tag{17}$$

where $\mathcal{D}_\delta(\mathbf{Z}) = \mathbf{U}_z R_\delta(\mathbf{\Sigma})\mathbf{V}_z^H$, $R_\delta(\mathbf{\Sigma}) = \text{diag}(\mathcal{S}(\sigma_i - \delta))$, $\mathcal{S}(x) = \max(x, 0)$ means the soft threshold function, $\mathbf{U}_z$ and $\mathbf{V}_z$ represent the orthogonal matrix after performing singular value decomposition for $\mathbf{Z}$, and $\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_r)$ indicates the singular value matrix. $\text{diag}(\cdot)$ and $(\cdot)^H$ denote the matrix diagonalization and conjugate transpose operation, respectively.

2) Update the Sparse Component **S***:* By fixing the variables **L**, **I**, and **U**, we obtain the minimization subproblem with respect to **S**, expressed as

$$\mathbf{S}^{k+1} = \underset{\mathbf{S}}{arg\min}\ \gamma_1 \|\mathbf{S}\|_1 + \frac{\rho}{2}\left\|\mathbf{Y} - \mathcal{G}\left\{\mathbf{L}^{k+1} + \mathbf{S}\right\} - \mathbf{I}^k + \mathbf{U}^k/\rho\right\|_F^2. \tag{18}$$

Similarly, the equation can also be linearized to obtain (19), which is shown at the bottom of this page, where κ_s indicates the linearization parameter for variable **S**. Then, the closed-form solution for the sparse subproblem can be derived as

$$\mathbf{S}^{k+1} = \mathcal{S}_{\frac{\gamma_1\kappa_s}{\rho}}\left(\mathbf{S}^k - \frac{\rho}{\kappa_s}\mathcal{H}\left\{\mathcal{G}\left\{\mathbf{L}^{k+1} + \mathbf{S}^k\right\} + \mathbf{I}^k - \mathbf{Y} + \frac{\mathbf{U}^k}{\rho}\right\}\right). \tag{20}$$

3) Update the Interference Component **I***:* By fixing the variables **L**, **S**, and **U**, we derive the minimization subproblem with respect to **I**, expressed as

$$\mathbf{I}^{k+1} = \underset{\mathbf{I}}{arg\min}\ \gamma_2 \|\mathbf{I}\|_* + \frac{\rho}{2}\left\|\mathbf{Y} - \mathcal{G}\left\{\mathbf{L}^{k+1} + \mathbf{S}^{k+1}\right\} - \mathbf{I} - \mathbf{U}^k/\rho\right\|_F^2. \tag{21}$$

The update expression for the interference component can be derived using the nuclear norm for proximal operator, which is presented as

$$\mathbf{I}^{k+1} = \mathcal{D}_{\gamma_2/\rho}\left(\mathbf{Y} - \mathcal{G}\left\{\mathbf{L}^{k+1} + \mathbf{S}^{k+1}\right\} - \mathbf{U}^k/\rho\right). \tag{22}$$

4) Update the Dual Component **U***:* By fixing the variables **L**, **S**, and **I**, the update expression for dual variable **U** is derived as

$$\mathbf{U}^{k+1} = \mathbf{U}^k + \rho\left(\mathbf{Y} - \mathcal{G}\left\{\mathbf{L}^{k+1} + \mathbf{S}^{k+1}\right\} - \mathbf{I}^{k+1}\right). \tag{23}$$

By iteratively updating the low rank, sparse, interference, and dual component according to (17), (20), (22), and (23), the algorithm proceeds until the update rate falls below a predefined threshold. Upon convergence, the iterative process terminates and outputs the final imaging result with RFI separated.

However, achieving effective simultaneous RFI separation and SAR imaging through this iterative algorithm faces two major challenges. First, it requires configuring multiple hyperparameters, including the sparsity/low-rank regularization coefficients, penalty factor, and linearization coefficients, whose manual tuning is notoriously difficult. Inappropriate settings can directly lead to failed RFI separation or degraded imaging quality. Second, the ADMM-based solution process typically requires a large number of iterations, resulting in prohibitive computational time that hinders its practical engineering application. To address these issues of parameter sensitivity and low computational efficiency, this article introduces a hyperparameter adaptation approach based on deep unfolding and gradient descent optimization.

B. SSI-Net

1) Network Architecture: Building upon the aforementioned ADMM iterative solution, this section constructs a corresponding deep unfolding network, named SSI-net. It is designed to leverage its data-driven capability for autonomously determining the required hyperparameters. The detailed architecture of SSI-net is illustrated in Fig. 6, where each layer incorporates four core modules: low-rank component update, sparse component update, RFI component update, and dual variable update. Notably, SSI-net takes the raw sampled echoes as input, and simultaneously outputs a high-quality focused SAR image along with the separated RFI, without requiring any additional preprocessing or postprocessing stages.

The ADMM solution not only requires predefined critical parameters but also suffers from slow convergence. To address these limitations, this section adopts a deep unfolding strategy that learns these parameters layer by layer. Denoting the parameters to be learned at each layer as $\Lambda^k = \{\gamma_1^k, \gamma_2^k, \rho^k, \kappa_l^k, \kappa_s^k\}$, this approach achieves rapid convergence with a limited number of network layers. When the total number of SSI-net layers is set to 10, the number of parameters requiring adaptive learning in the network amounts to 50.

On the other hand, it is necessary to impose nonnegative constraints on these parameters to be learned, which is constrained as

$$softplus\left(\Lambda^k\right) = \log\left(1 + e^{\Lambda^k}\right). \tag{24}$$

2) Dataset Construction: To ensure stable network training, this section constructs a dataset based on the high-resolution SAR complex image released by the Sandia Laboratory. First, the original complex-valued images are cropped to obtain 3100

$$\mathbf{L}^{k+1} = \underset{\mathbf{L}}{arg\min}\ \|\mathbf{L}\|_* + \frac{\kappa_l}{2}\left\|\mathbf{L} - \mathbf{L}^k + \frac{\rho}{\kappa_l}\mathcal{H}\left\{\mathcal{G}\left\{\mathbf{L}^k + \mathbf{S}^k\right\} + \mathbf{I}^k - \mathbf{Y} + \frac{\mathbf{U}^k}{\rho}\right\}\right\|_F^2 \tag{16}$$

$$\mathbf{S}^{k+1} = \underset{\mathbf{S}}{arg\min}\ \gamma_1 \|\mathbf{S}\|_1 + \frac{\kappa_s}{2}\left\|\mathbf{S} - \mathbf{S}^k + \frac{\rho}{\kappa_s}\mathcal{H}\left\{\mathcal{G}\left\{\mathbf{L}^{k+1} + \mathbf{S}^k\right\} + \mathbf{I}^k - \mathbf{Y} + \frac{\mathbf{U}^k}{\rho}\right\}\right\|_F^2 \tag{19}$$

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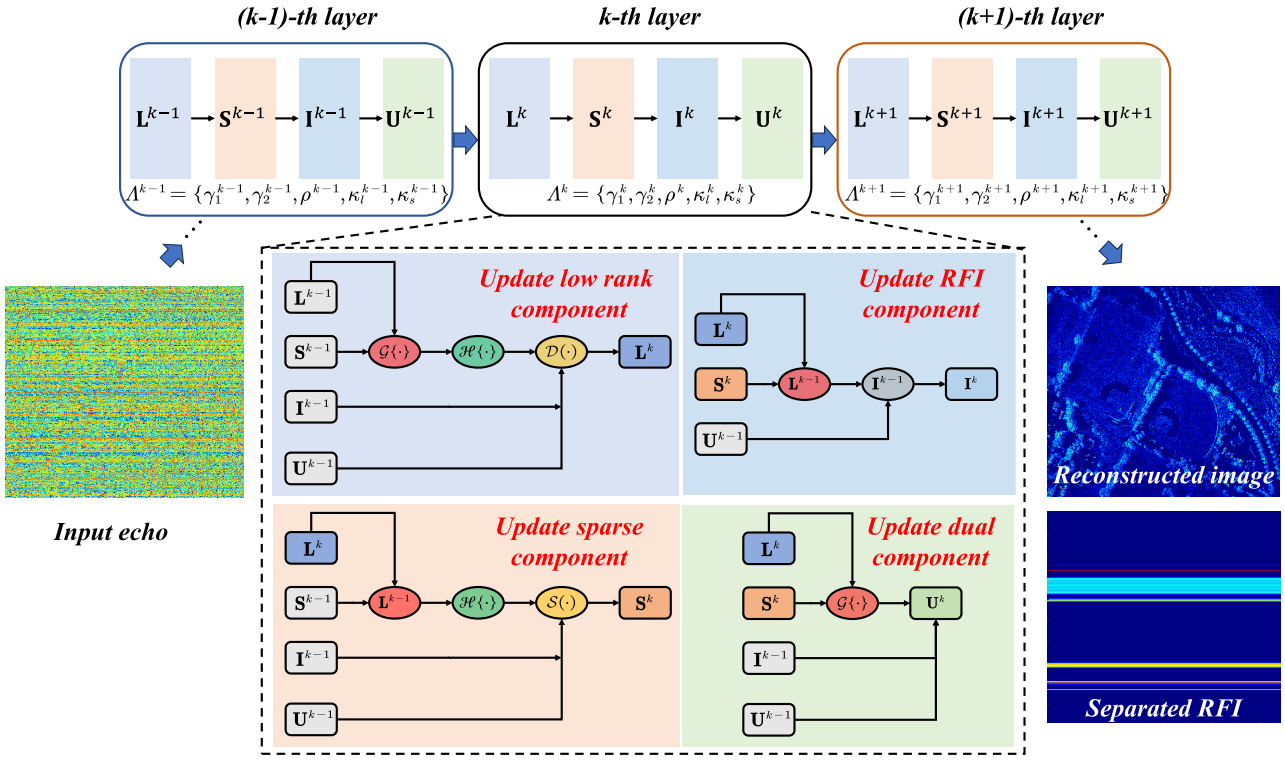

Fig. 6. SSI-net architecture.

TABLE II
RFI PARAMETERS FOR DATASET CONSTRUCTION

RFI types	Symbol	Value
$I_{nbi}(t)$	K	[1, 4]
	$f_{n,k}$	[9.3, 9.9]GHz
	φ_k	[0, 2π]
$I^l_{wbi}(t)$	K	[1, 3]
	$f_{w,k}$	[9.4, 9.8]GHz
	γ_k	[1×10^{13}, 4×10^{14}]
	φ_k	[0, 2π]
$I^s_{wbi}(t)$	K	[1, 3]
	$f_{w,k}$	[9.4, 9.8]GHz
	α_k	[10, 100]
	$f^s_{w,k}$	[0.1, 5]MHz
	φ_k	[0, 2π]

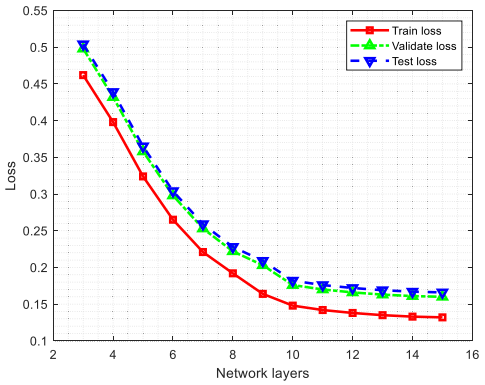

Fig. 7. Curves of training, validation, and test losses as functions of the number of network layers.

images of size 512 × 512, denoted as $\mathbf{X}_{label}$. Subsequently, the echo signal corresponding to each image is generated through the inverse imaging operator $\mathcal{G}\{\cdot\}$, denoted as $\mathbf{Y}_E$. Besides, for the RFI sources, three–ten interference signals are randomly generated according to the parameters listed in Table II, denoted as $\mathbf{I}_{label}$, which are then superimposed onto the simulated echoes.

To enhance the dataset and improve the model's generalization capability, we augment the data by superimposing two distinct types of RFI signals onto each simulated echo, resulting in a total of 6200 samples. The specific data split is as follows: 6000 samples are allocated for training, 100 for validation, and an additional 100 are reserved for testing.

To evaluate the performance of SSI-net in simultaneous imaging and RFI separation, a loss function must be defined to quantify the discrepancy among the reconstructed image, the separated RFI, and their corresponding ground truths. The loss function is formulated as

$$Loss = \frac{1}{N}\left\{\sum_{i=1}^{N}\frac{\|\mathbf{X}_r-\mathbf{X}_{label}\|_F^2}{\|\mathbf{X}_{label}\|_F^2}+\alpha\sum_{i=1}^{N}\frac{\|\mathbf{I}_r-\mathbf{I}_{label}\|_F^2}{\|\mathbf{I}_{label}\|_F^2}\right\} \quad (25)$$

where $\mathbf{X}_r = \mathbf{L}_r + \mathbf{S}_r$ and $\mathbf{I}_r$ represent the image after network reconstruction and the separated interference, respectively. α indicates the weight coefficient, which is set to 0.2 in the network training procedure. N means the batch number.

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The network training procedure was implemented on a platform with a NVIDIA GeForce RTX3080 (20 GB), an Intel[1] Core[2] Ultra 7 CPU, two RAMs (96 GB), and a Windows 11 operating system. The number of network layers in SSI-net is set to 10. When determining the layer number, we trained the network with varying layer depths and plotted the curves of training loss, validation loss, and test loss as functions of the number of network layers, which is shown in Fig. 7.

The red, green, and blue curves in Fig. 7 represent the training loss, validation loss, and test loss as functions of the number of network layers, respectively. It can be observed that when the number of network layers is less than 9, all three loss curves continuously decrease. When the number of layers is greater than or equal to 9, although each loss curve still exhibits a decreasing trend, the descending gradient becomes relatively gentle. However, as the number of network layers keeps increasing, the time required to train the same number of epochs also increases. Therefore, to strike a balance between training accuracy and computational cost, the number of network layers was set to 10.

C. Gradient Descent Optimization Method

Algorithm 1 Hyperparameter Design Method Based on Gradient Descent Optimization

1: Initialize learnable hyperparameters $\Lambda^1, \Lambda^2, \ldots, \Lambda^K$ using optimizer, and set learning rate to 0.001.
2: Initialize $\mathbf{I}^0$, $\mathbf{S}^0$, $\mathbf{I}^0$, $\mathbf{U}^0$, and set initial iteration count $k = 1$ and $t = 1$.
3: **while** $t \leq T$ **do**
1) **while** $k \leq K$ **do**
a) Update low-rank component $\mathbf{L}^k$ using SVT algorithm according to (17).
b) Update sparse component $\mathbf{S}^k$ using soft thresholding operation according to (20).
c) Update interference component $\mathbf{I}^k$ using SVT algorithm similarly according to (22).
d) Update dual-variable $\mathbf{U}^k$ according to (23).
e) $k = k + 1$.
2) **end while**
3) Output $\mathbf{X}^K = \mathbf{L}^K + \mathbf{S}^K$ and $\mathbf{I}^K$.
4) Calculate loss function.
5) Backward propagation of loss.
6) Optimizer steps.
7) if loss < preset threshold, break.
8) $t = t + 1$.
4: **end while**
5: Output $\Lambda^1, \Lambda^2, \ldots, \Lambda^K$, $\mathbf{X}$ and $\mathbf{I}$.

While the SSI-net constructed via deep unfolding in Section III-B can learn hyperparameters from data to achieve simultaneous RFI separation and SAR imaging, its performance is contingent on the availability of substantial training samples. To address scenarios with limited or no training data, this section proposes a gradient-descent-based hyperparameter optimization method, which is summarized in Algorithm 1.

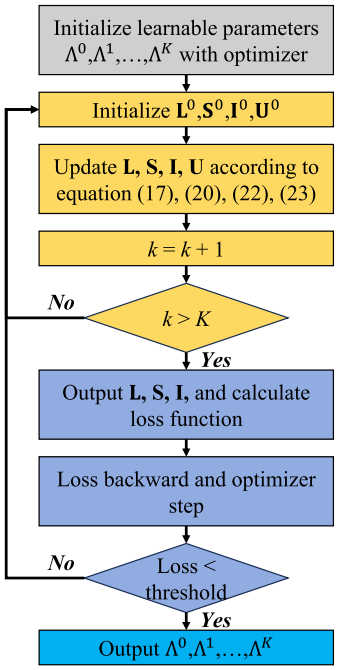

Fig. 8. Flowchart of the gradient descent optimization method.

TABLE III
SIMULATION SYSTEM PARAMETERS

Parameter	Value
Signal Carrier Frequency	9.6GHz
Signal Bandwidth	500MHz
Sampling Frequency	800MHz
Pulse Width	5us
Pulse Repetition Frequency	1000Hz
Platform Position	(5000,0,5000)m
Platform Velocity	150m/s

If a small set of training samples is available, the loss function can be computed according to (25). Conversely, in the complete absence of training samples, metrics such as image entropy or the reciprocal of image sharpness could serve as alternative loss functions. To present the algorithm more intuitively, Fig. 8 provides a flowchart of the algorithm's operation.

For the optimizer selection, this article employs the built-in Adam optimizer from PyTorch to perform gradient descent. It should be noted that the method remains compatible with alternative optimizers such as stochastic gradient descent (SGD) and limited-memory Broyden-Fletcher-Goldfarb-Shanno (LBFGS). Within this gradient descent optimization framework, we can automatically obtain a set of hyperparameters that satisfy the preset loss requirement. Thus, the final objective of simultaneously RFI suppression and high-quality SAR imaging can be achieved.

IV. Numerical Simulation and Experiment

This section validates the effectiveness of the proposed simultaneous RFI separation and SAR imaging framework

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through four experiments. Experiment A demonstrates the framework's performance under a narrowband RFI scenario, while experiment B addresses hybrid RFI scenarios. The system parameters used are presented in Table III. Experiments C and D involve specific airborne flight experiments, in which the superior RFI suppression performance of the proposed framework is further compared and analyzed.

A. Narrowband RFI Simulation Experiment

In this simulation experiment, three NBIs are configured with center frequencies at 9.5, 9.6, and 9.7 GHz, and initial phases of 0, 0.5π, and π, respectively. The interference-to-signal ratio (ISR) is set to 10 dB. For a comprehensive performance evaluation, this section selects several representative methods for comparison. Their core principles and inherent limitations are thoroughly analyzed as follows.

1) *FD [21]:* This method operates by designing matched filters in the FD to suppress RFI. However, it inevitably attenuates target echoes within the filtered frequency bands, leading to degraded resolution and elevated sidelobes in the imaging results.
2) *TFD [22]:* Compared with the FD method, this approach achieves lower frequency resolution. Consequently, suppressing NBI causes more significant loss of echo bandwidth, resulting in more severe resolution degradation and higher sidelobes than those of the FD method.
3) *RPCA [24]:* This technique relies on the assumption that target echoes are sparse and RFI has low-rank characteristics. However, in complex scenes, the sparsity of SAR target echoes is often insufficient, leading to model mismatch. Its imaging performance is inferior to that of filtering-based methods.
4) *FELS-net [33]:* The core idea is to introduce two learnable nonlinear transformations to enhance the sparsity of target echoes and the low-rank property of RFI, thereby achieving separation based on the RPCA principle. Nevertheless, the processing results exhibit noticeable loss of image details, indicating that the network tends to excessively suppress weak target echoes.
5) *IMN-net [32]:* This method shows poor adaptability to different types and power levels of RFI, often failing to achieve sufficient RFI suppression. Significant residual interference remains in the output images.
6) *2DSA [35]:* Its performance is sensitive to the interference type. It struggles to achieve satisfactory suppression performance for NBI and SFM WBI.
7) *BSF [36]:* Although capable of handling various RFI types, it inevitably causes amplitude loss in strong point scatterers during the suppression procedure.

The imaging results after RFI suppression using different methods are shown in Fig. 9. The results are organized rowwise: the first row displays the images before suppression and after FD, TFD, and RPCA processing. The second row presents the results after FELS-net, IMN-net, 2DSA, and BSF processing. The third row shows the results of ADMM with random hyperparameters (denoted as Rand-ADMM for simplicity), ADMM with hyperparameters optimized by gradient descent using the Adam optimizer (denoted as Adam-ADMM), SSI-net, and the RFI-free image.

Besides, it is noteworthy that the 2DSA and BSF methods operate directly on the RFI contaminated images, categorizing them as postprocessing approaches, whereas all other compared methods belong to the preprocessing category.

The configuration of hyperparameters is crucial to the performance of the proposed framework. When these hyperparameters are set inappropriately, the iterative solution fails to achieve the expected performance, leaving significant residual interference in the imaging results, as shown in Fig. 9(i). In contrast, gradient descent optimization using the Adam optimizer can automatically obtain effective hyperparameters, and the SSI-net can learn these parameters through training. Both strategies enable excellent simultaneous RFI separation and imaging performance, as demonstrated in Fig. 9(j) and (k).

For a quantitative evaluation of the RFI suppression performance across different methods, this article employs three metrics: the normalized mean square error (NMSE), the structural similarity index (SSIM), and the peak signal-to-noise ratio (PSNR). Their definitions are presented as

$$NMSE = \frac{\|\mathbf{X}_r - \mathbf{X}_{label}\|_F^2}{\|\mathbf{X}_{label}\|_F^2} \tag{26}$$

$$PSNR = 20\log\left(\frac{\sqrt{N_a N_r}\max\left(|\mathbf{X}_r|\right)}{\|\mathbf{X}_r - \mathbf{X}_{label}\|_F}\right) \tag{27}$$

$$SSIM = \frac{(2\mu_1\mu_2 + c_1)(2\sigma_{12} + c_2)}{\left(\mu_1^2 + \mu_2^2 + c_1\right)\left(\sigma_1^2 + \sigma_2^2 + c_2\right)} \tag{28}$$

where N_a and N_r represent the sampling numbers of azimuth and range dimensions, respectively. μ_1, μ_2, σ_1, and σ_2 indicate the mean and standard deviation of both recovered image $\mathbf{X}_r$ and label image $\mathbf{X}_{label}$, respectively. σ_{12} denotes the covariance between the two images. $c_1 = E^2 \times 10^{-4}$, $c_2 = 9E^2 \times 10^{-4}$, and E means the dynamic range of imaging results.

The quantitative evaluation results for SAR images reconstructed by different methods are summarized in Table IV, comparing performance at ISR levels of 5, 10, and 15 dB (all metrics averaged over 100 Monte Carlo experiments). It can be observed that the proposed SSI-net and Adam-ADMM methods consistently achieve superior image quality after RFI suppression compared to all other methods. Notably, SSI-net attains the best scores across all metrics. Although Adam-ADMM performs slightly below SSI-net, its reconstruction quality remains fully adequate for demanding downstream applications, such as target recognition and feature extraction. Crucially, Adam-ADMM offers a significant practical advantage by requiring no network pretraining, making it highly suitable for scenarios with limited or no training samples.

Specifically, the FD, TFD, RPCA, 2DSA, and BSF methods do not require training samples to extract prior information and can be directly applied to process SAR echoes or images contaminated by RFI. By comparing with these methods in terms of both qualitative imaging results and quantitative metrics such as NMSE, PSNR, and SSIM, the superior performance of the Adam-ADMM method for simultaneous RFI separation and SAR imaging under limited or no training samples conditions is demonstrated to a certain extent. In

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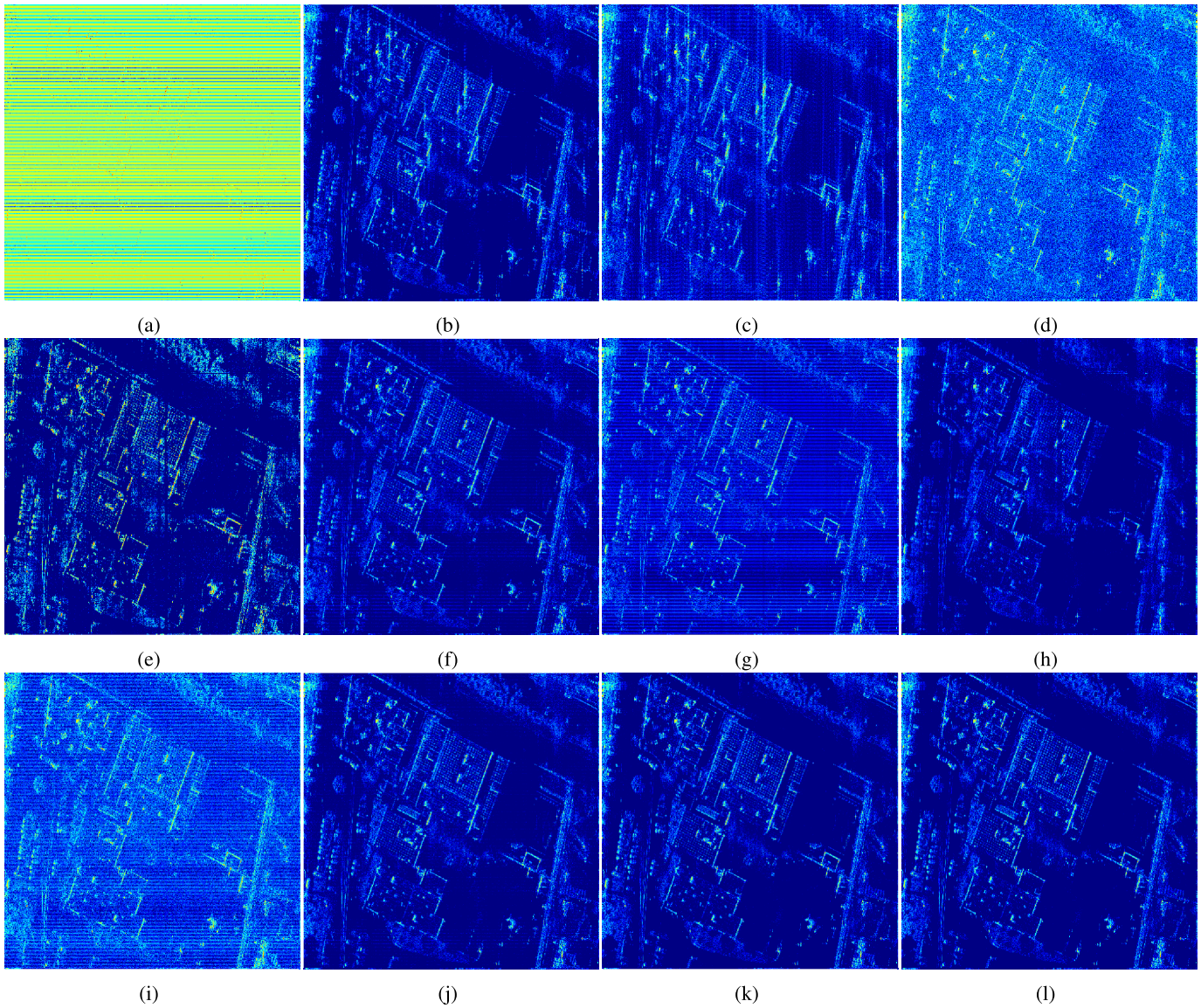

Fig. 9. Imaging results of experiment 1. (a) Before RFI suppression. (b) FD [21]. (c) TFD [22]. (d) RPCA [24]. (e) FELS-net [33]. (f) IMN-net [32]. (g) 2DSA [35]. (h) BSF [36]. (i) Rand-ADMM. (j) Adam-ADMM. (k) SSI-net. (l) RFI-free result.

TABLE IV
PERFORMANCE OF DIFFERENT METHODS FOR EXPERIMENT A

ISR(dB)	Metrics	Before	FD	TFD	RPCA	FELS-net	IMN-net	2DSA	BSF	Rand-ADMM	Adam-ADMM	SSI-net
5	NMSE	2.1472	0.2124	0.5073	0.7161	0.3415	0.2206	0.4946	0.3086	0.5613	**0.1766**	**0.0619**
	SSIM	0.4330	0.9941	0.9412	0.6522	0.9624	0.9921	0.9372	0.9943	0.8587	**0.9951**	**0.9997**
	PSNR	34.6840	56.0215	46.9004	38.3825	41.6455	56.2418	48.6159	52.5054	42.6505	**58.1632**	**67.0862**
10	NMSE	3.9740	0.2592	0.5585	0.7363	0.5682	0.2519	0.5187	0.5315	0.6125	**0.1856**	**0.0955**
	SSIM	0.3112	0.9916	0.9166	0.6223	0.9091	0.9876	0.9287	0.9822	0.7389	**0.9944**	**0.9991**
	PSNR	28.9033	53.9960	45.8619	37.9255	36.6303	55.0161	48.1828	47.4490	40.3248	**57.6828**	**63.5583**
15	NMSE	7.7636	0.4801	0.5591	0.7823	0.7532	0.2521	0.5791	0.7787	0.8789	**0.1991**	**0.1200**
	SSIM	0.0775	0.9734	0.8945	0.5413	0.8750	0.9877	0.9120	0.9626	0.4283	**0.9931**	**0.9937**
	PSNR	23.2534	47.2140	45.0548	36.5955	34.5335	54.9787	47.1149	43.6910	34.5133	**57.0820**	**59.9338**

contrast, FELS-net and IMN-net represent model-driven and data-driven deep-learning-based RFI suppression methods, respectively, and are highly typical. We retrained these two networks using a small number of samples to evaluate their RFI suppression performance with limited samples. Using an interval of 1000 training samples, we trained the corresponding network models with sample sizes of 1000, 2000, 3000, 4000, 5000, and 6000, respectively. For the NBI experiment,

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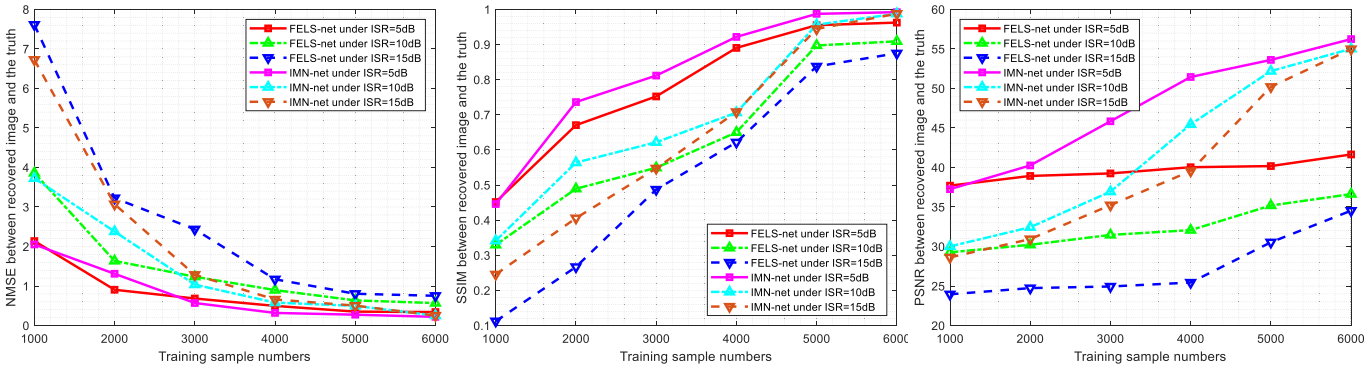

Fig. 10. Curves of various performance metrics of SAR imaging results as a function of the number of training samples in the NBI experiment. (a) NMSE. (b) SSIM. (c) PSNR.

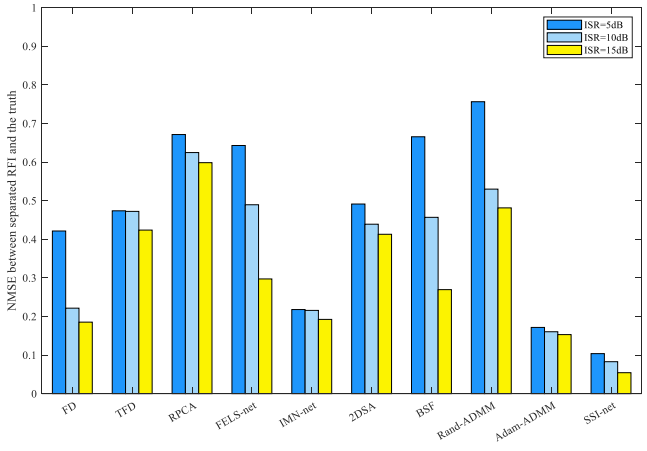

Fig. 11. Bar chart of the NMSE between the RFI separated by different methods and the truth in experiment A.

Fig. 10 presents the curves of SAR imaging performance metrics as a function of the number of training samples.

It is worth noting that each performance metric in Fig. 10 represents the average over 100 test samples. It can be observed that as the number of training samples increases, the NMSE between the SAR imaging results and the truth gradually decreases, while both SSIM and PSNR gradually increase. Furthermore, when the number of training samples exceeds 5000, the changes in various performance metrics gradually level off. Therefore, setting the number of training samples to 6000 for each network in this article serves two purposes: on the one hand, it ensures control of variables; on the other hand, it achieves a favorable balance between training time and accuracy.

Similar to the evaluation of SAR imaging performance, we also conducted Monte Carlo experiments with 100 trials to compute the NMSE between the RFI separated by different methods and the truth. For a more intuitive comparison of RFI separation performance among different methods, Fig. 11 presents a bar chart of the NMSE between the separated RFI and the truth, from which two conclusions can be drawn: 1) as the ISR increases, the NMSE between the RFI separated by different methods and the true RFI gradually decreases and 2) compared with other methods, SSI-net and Adam-ADMM exhibit superior RFI separation performance. Specifically, the NMSE between the RFI separated by Adam-ADMM and the true RFI does not exceed 0.18, while that of SSI-net does not exceed 0.11.

B. Hybrid RFI Simulation Experiment

For this simulation experiment, a hybrid scenario comprising seven interference sources is constructed, including three NBIs with center frequencies at 9.53, 9.61, and 9.77 GHz; two wideband LFM interferences with center frequencies of 9.725 and 9.425 GHz and corresponding chirp rates of 1.3×10^{13} and 1.5×10^{14}; and two wideband SFM interferences with center frequencies of 9.6 and 9.8 GHz, modulation indices of 30 and 20, and modulation frequencies of 1 and 0.5 MHz, respectively. The initial phases of all interference sources are randomly set within the range $[0, 2\pi]$. At an ISR of 10 dB, the imaging results after RFI suppression using different methods are shown in Fig. 12.

Analysis of the imaging results reveals distinct limitations in existing methods. The FD and TFD methods cause noticeable sidelobe elevation and resolution degradation. Specifically, the FD method indiscriminately filters both interference and target signals, incurring more severe bandwidth loss than the TFD method. The RPCA method suffers from poor reconstruction due to model mismatch, while FELS-net leads to significant loss of image details. IMN-net exhibits incomplete suppression with residual interference, and 2DSA is effective only against LFM WBI, failing under hybrid scenarios. Although BSF adapts well to various interference types, it introduces amplitude loss in strong scatterers, thereby hindering image interpretation. In contrast, the proposed Adam-ADMM and SSI-net methods successfully separate diverse interference types while reconstructing high-quality SAR imagery.

Under this hybrid RFI scenario, Table V presents the quantitative metrics for SAR images reconstructed by various methods under different ISR conditions. The results demonstrate that the proposed Adam-ADMM and SSI-net methods achieve superior image quality alongside effective interference suppression, confirming their robustness and effectiveness. Notably, while SSI-net holds an overall advantage in NMSE

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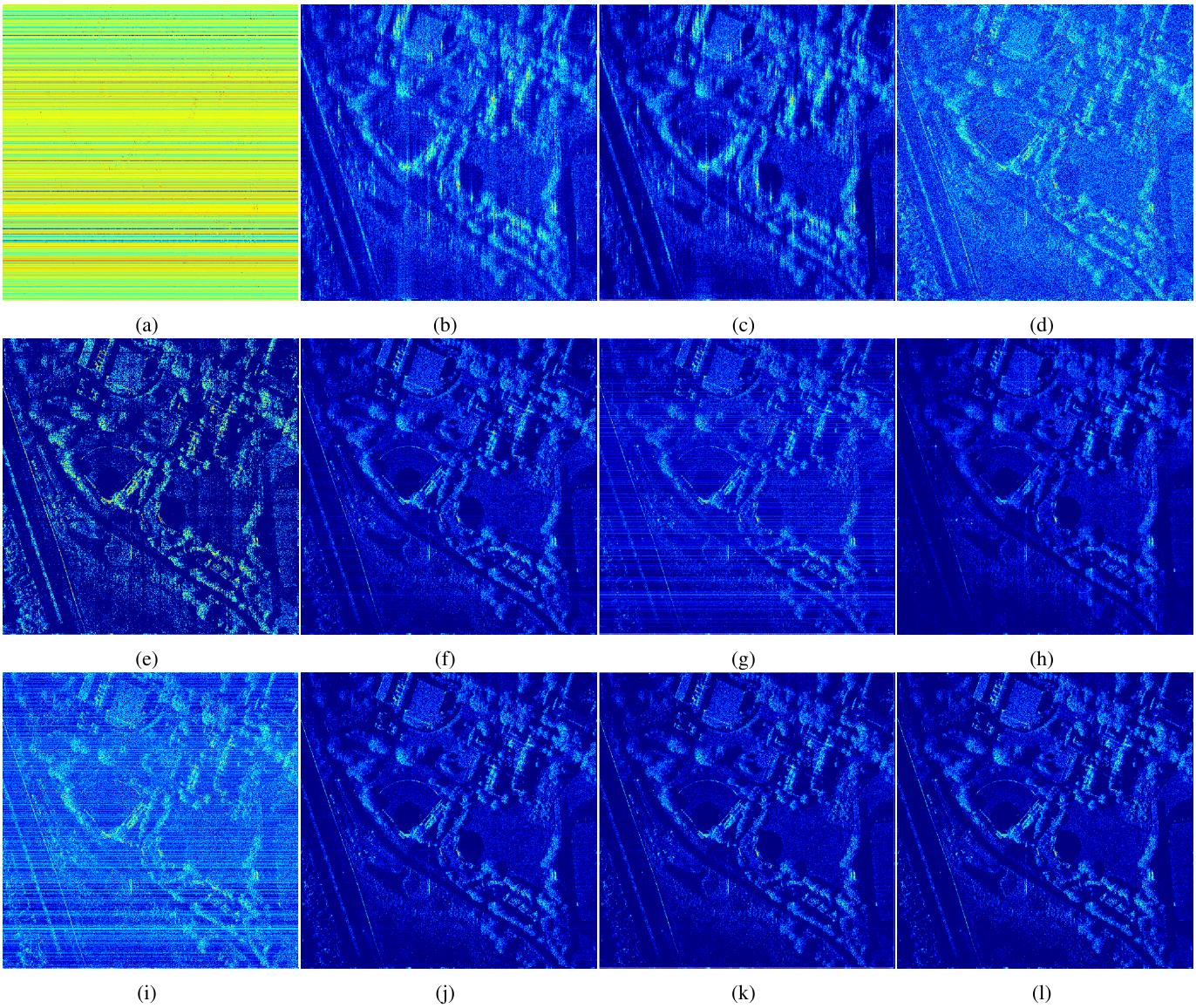

Fig. 12. Imaging results of experiment 2. (a) Before RFI suppression. (b) FD [21]. (c) TFD [22]. (d) RPCA [24]. (e) FELS-net [33]. (f) IMN-net [32]. (g) 2DSA [35]. (h) BSF [36]. (i) Rand-ADMM. (j) Adam-ADMM. (k) SSI-net. (l) RFI-free result.

TABLE V
PERFORMANCE OF DIFFERENT METHODS FOR EXPERIMENT B

ISR(dB)	Metrics	Before	FD	TFD	RPCA	FELS-net	IMN-net	2DSA	BSF	Rand-ADMM	Adam-ADMM	SSI-net
5	NMSE	4.7267	0.6865	0.6305	0.6210	0.5122	0.2696	0.5395	0.4100	0.5674	**0.2067**	**0.1985**
	SSIM	0.2478	0.9195	0.9635	0.8102	0.8726	0.9920	0.9646	0.9910	0.8535	**0.9975**	**0.9970**
	PSNR	29.1545	44.1963	46.3773	41.6945	37.5727	55.9284	50.2600	51.6448	43.1004	**58.6009**	**59.1334**
10	NMSE	12.2484	0.7629	0.7282	0.7254	0.6412	0.2698	0.5720	0.4572	0.6741	**0.2189**	**0.1856**
	SSIM	0.0502	0.8807	0.9325	0.7158	0.8953	0.9918	0.9388	0.9888	0.7206	**0.9972**	**0.9973**
	PSNR	20.3871	42.0653	43.7951	39.5478	37.6666	55.8808	49.7143	50.6553	39.8689	**58.2244**	**58.3504**
15	NMSE	37.9176	0.8590	0.7917	0.6261	0.6796	0.7193	0.6027	0.3954	1.3049	**0.2430**	**0.2256**
	SSIM	0.0027	0.7623	0.8947	0.6130	0.8844	0.7321	0.9527	0.9745	0.3409	**0.9963**	**0.9959**
	PSNR	9.9000	39.4288	41.8177	35.4974	37.0675	30.6162	49.3220	48.1123	31.7246	**57.4230**	**58.0159**

and PSNR, it slightly underperforms Adam-ADMM in terms of SSIM under both 5- and 15-dB ISR conditions.

In addition, we also conducted simulation validations under the hybrid interference experiment to examine the variation of SAR imaging performance metrics with respect to the number of training samples. The results are shown in Fig. 13.

These results indicate that after RFI suppression by FELS-net and IMN-net, as the number of training samples

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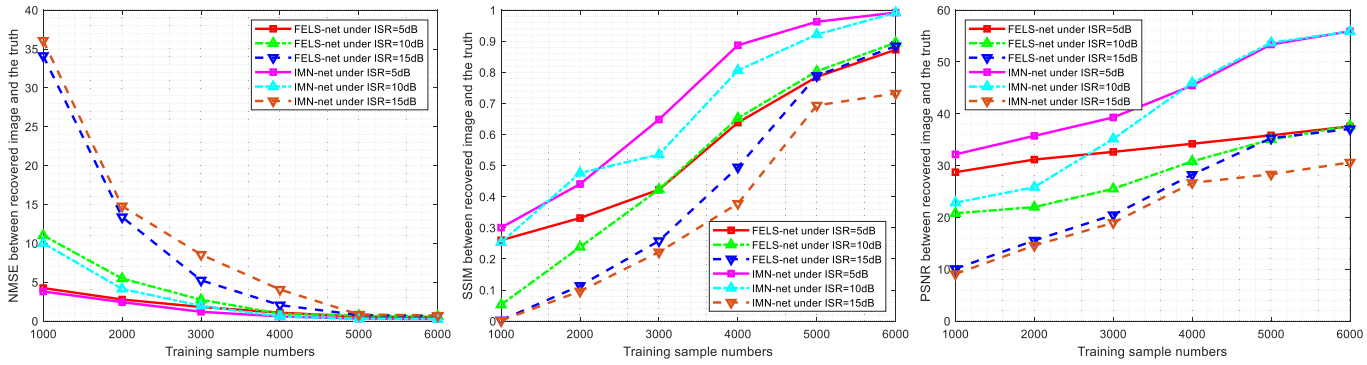

Fig. 13. Curves of various performance metrics of SAR imaging results as a function of the number of training samples in the hybrid interference experiment. (a) NMSE. (b) SSIM. (c) PSNR.

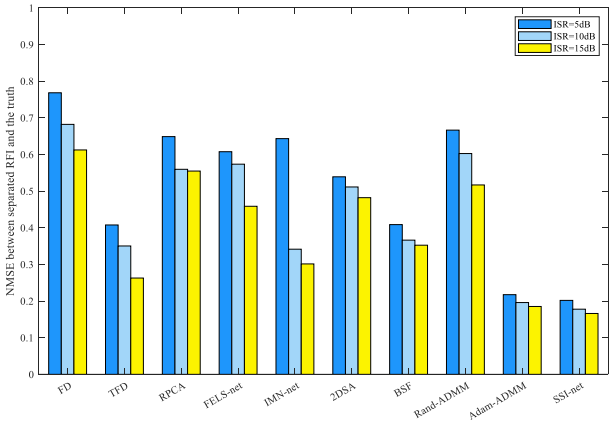

Fig. 14. Bar chart of the NMSE between the RFI separated by different methods and the truth in experiment B.

TABLE VI
SYSTEM PARAMETERS IN YINCHUAN

Parameter	Value
Signal Carrier Frequency	X band
Signal Bandwidth	400MHz
Sampling Frequency	800MHz
Pulse Width	8us
Pulse Repetition Frequency	3200Hz
Platform Position	(1652.83,499.48,2153.52)m
Platform Velocity	58.76m/s

increases, the NMSE between the SAR imaging results and the truth gradually decreases, while both SSIM and PSNR gradually increase. Consequently, in few sample scenarios, the RFI suppression performance further deteriorates. As one of the innovations presented in this article, Adam-ADMM is based on the established hybrid domain joint low-rank and sparse-constrained model. It leverages gradient descent to adaptively determine the various hyperparameters involved in the solution model, and its performance does not depend on the number of training samples. Through the above comparative experiments and analysis results, the RFI suppression performance of the proposed algorithm can be quantitatively evaluated.

Similarly, to more intuitively demonstrate the RFI separation performance of different methods, Fig. 14 presents a bar chart of the NMSE between the separated RFI and the true values. As can be observed, as the ISR increases, the NMSE between the RFI separated by different methods and the true RFI gradually decreases. Moreover, compared with the other methods, both SSI-net and Adam-ADMM exhibit superior RFI separation performance.

C. SAR Flight Experiment in Yinchuan

In October 2020, our research group conducted an airborne SAR flight experiment in Yinchuan, Ningxia, China. The experiment utilized a Cessna-208 aircraft equipped with an X-band SAR system. The system operated with a bandwidth of 400 MHz, a pulsewidth of 8 μs, a pulse repetition frequency (PRF) of 3200 Hz, and a sampling rate of 800 MHz. The aircraft maintained an altitude of 2153.52 m and a velocity of 58.76 m/s during the flight experiment. Detailed system parameters are summarized in Table VI.

This flight experiment conducted observations of the same scene over two consecutive days. During the first day's flight, persistent RFI was present in the scene. Echoes at three different azimuth moments were selected and subjected to time–frequency analysis, the results of which are shown in Fig. 15. It can be observed that the echoes at all three selected moments contain interference with different chirp characteristics. This interference obscures nearly half of the SAR image, as shown in Fig. 16(a). In contrast, no RFI was observed during the second day's flight, thus providing a clear reference image, as shown in Fig. 16(l). This controlled dataset offers strong support for validating the effectiveness of the proposed framework. Finally, Fig. 16(b)–(k) shows the processing results of the RFI contaminated data using different methods.

As shown in Fig. 16, the imaging results processed by the FD, TFD, Adam-ADMM, and SSI-net methods exhibit almost no residual RFI. In contrast, the RPCA method fails to achieve the expected separation performance due to model mismatch, and the FELS-net method leads to severe loss of image detail. Meanwhile, discernible residual RFI persists in the results processed by the IMN-net, 2DSA, and BSF methods. The

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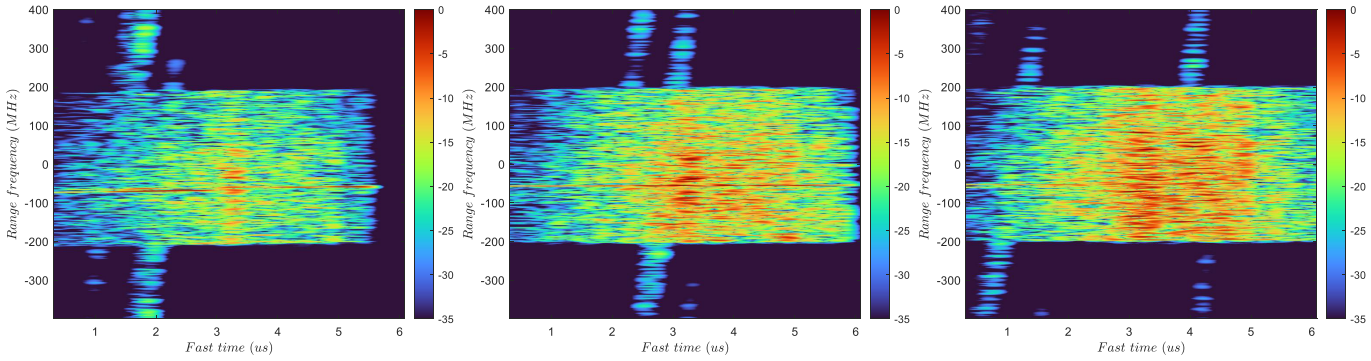

Fig. 15. Echo in the TFD for Yinchuan flight experiment. (a) Moment 1. (b) Moment 2. (c) Moment 3.

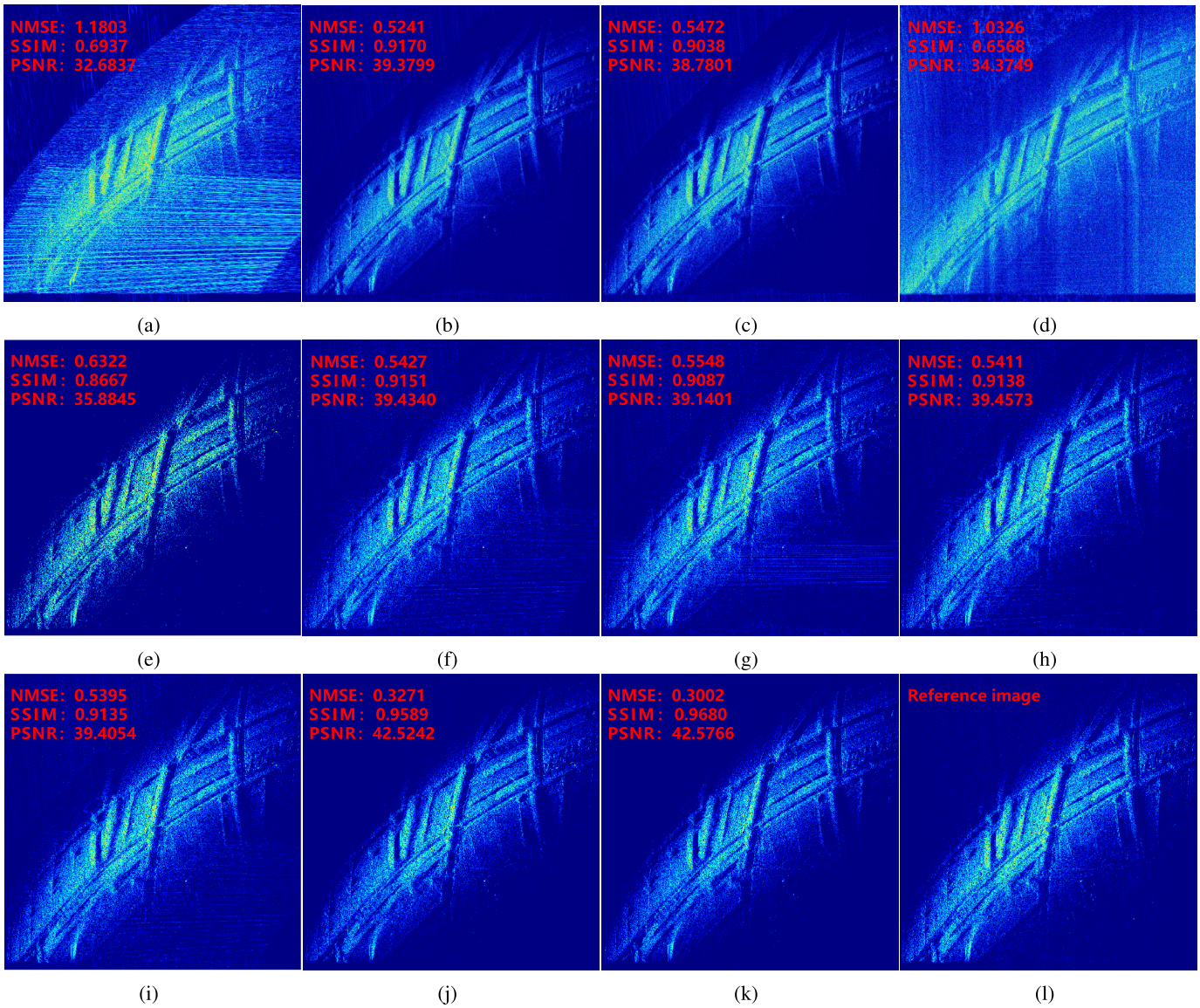

Fig. 16. Imaging results of flight experiment in Yinchuan. (a) Imaging result of the first day. (b) FD [21]. (c) TFD [22]. (d) RPCA [24]. (e) FELS-net [33]. (f) IMN-net [32]. (g) 2DSA [35]. (h) BSF [36]. (i) Rand-ADMM. (j) Adam-ADMM. (k) SSI-net. (l) Imaging result of the second day.

Rand-ADMM method also shows residual RFI in its output, which is attributed to suboptimal hyperparameter settings.

For quantitative evaluation, the RFI free image from the second day serves as the reference for calculating the NMSE,

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TABLE VII
PERFORMANCE OF DIFFERENT METHODS FOR YINCHUAN FLIGHT EXPERIMENT

Metrics	Before	FD	TFD	RPCA	FELS-net	IMN-net	2DSA	BSF	Rand-ADMM	Adam-ADMM	SSI-net
NMSE	1.1803	0.5241	0.5472	1.0326	0.6322	0.5427	0.5548	0.5411	0.5395	**0.3271**	**0.3002**
SSIM	0.6937	0.9170	0.9038	0.6568	0.8667	0.9151	0.9087	0.9138	0.9135	**0.9589**	**0.9680**
PSNR	32.6837	39.3799	38.7801	34.3749	35.8845	39.4340	39.1401	39.4573	39.4054	**42.5242**	**42.5766**

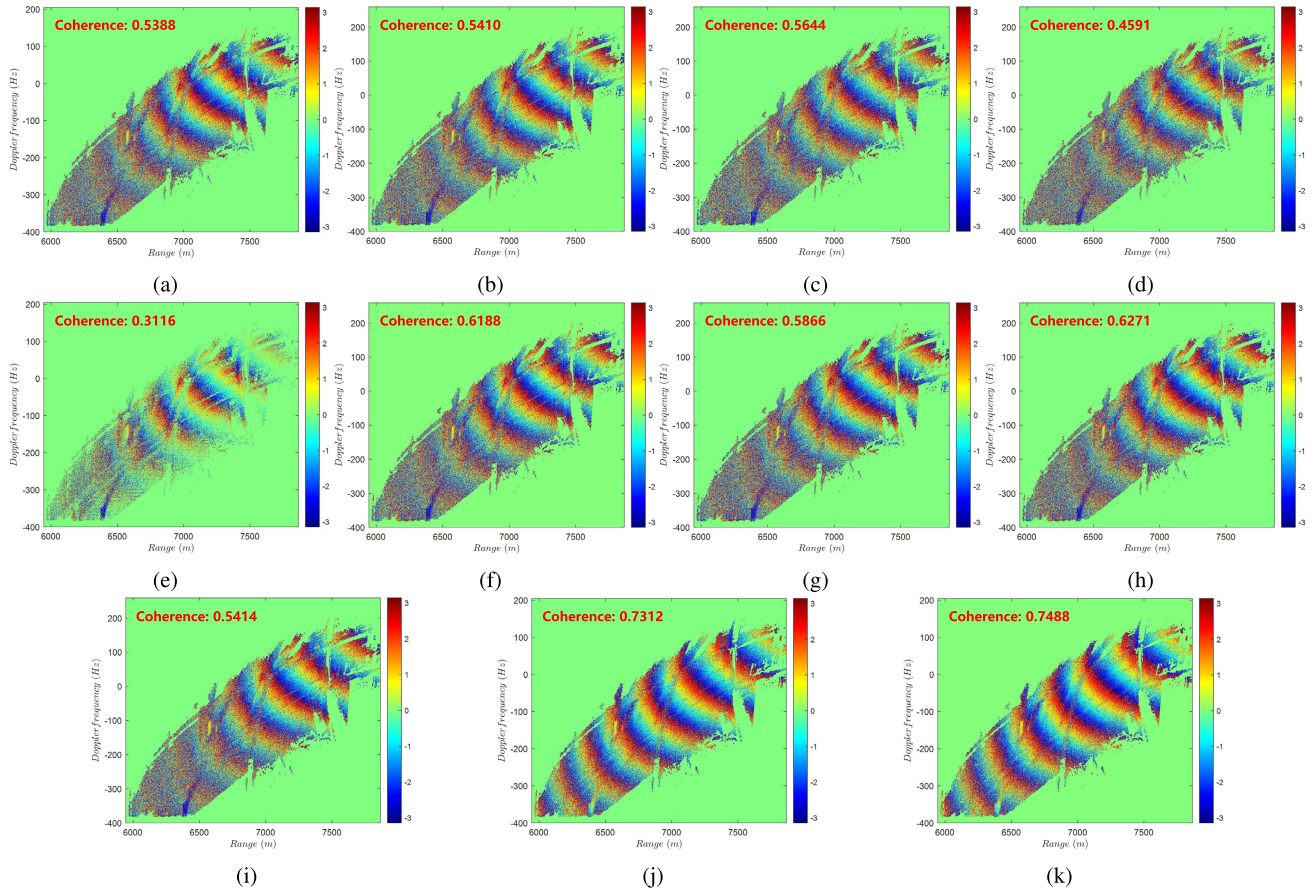

Fig. 17. Interferograms between the RFI suppressed results by different methods and the reference image in Yinchuan. (a) Before RFI suppression. (b) FD [21]. (c) TFD [22]. (d) RPCA [24]. (e) FELS-net [33]. (f) IMN-net [32]. (g) 2DSA [35]. (h) BSF [36]. (i) Rand-ADMM. (j) Adam-ADMM. (k) SSI-net.

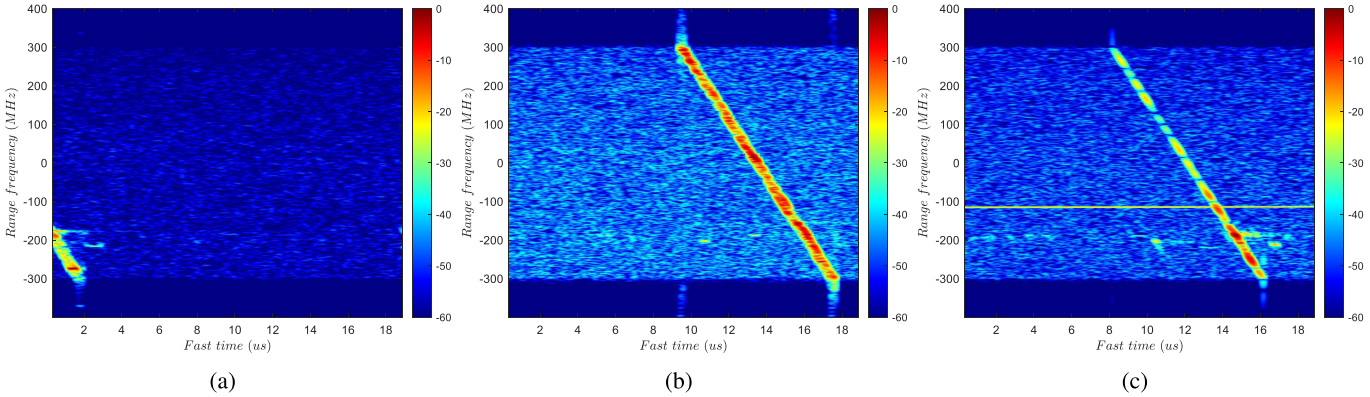

Fig. 18. Installation site of the flight experiment in Yantai.

SSIM, and PSNR metrics. These metrics are computed for the original contaminated data and for the output of each method, as shown in Table VII. The results indicate that Adam-ADMM and SSI-net achieve the best quantitative

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TABLE VIII
SYSTEM PARAMETERS IN YANTAI

Parameter	Value
Signal Carrier Frequency	X band
Signal Bandwidth	600MHz
Sampling Frequency	800MHz
Pulse Width	8us
Pulse Repetition Frequency	3200Hz
Platform Position	(2712.51,373.22,3184.53)m
Platform Velocity	75.24m/s

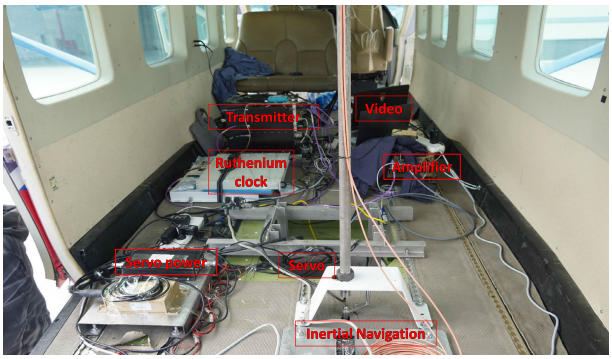

Fig. 19. Echo in the TFD for Yantai flight experiment. (a) Moment 1. (b) Moment 2. (c) Moment 3.

scores. Specifically, compared with the FD, TFD, RPCA, FELS-net, IMN-net, 2DSA, BSF, and Rand-ADMM methods, Adam-ADMM reduces the NMSE by 0.1970, 0.2201, 0.7055, 0.3051, 0.2156, 0.2277, 0.2140, and 0.2124, respectively. It improves the SSIM by 0.0419, 0.0551, 0.3021, 0.0922, 0.0438, 0.0502, 0.0451, and 0.0454, and increases the PSNR by 3.1443, 3.7441, 8.1493, 6.6397, 3.0902, 3.3841, 3.0669, and 3.1188, respectively. Similarly, SSI-net demonstrates NMSE reductions of 0.2239, 0.2470, 0.7324, 0.3320, 0.2425, 0.2546, 0.2409, and 0.2393; SSIM gains of 0.0510, 0.0642, 0.3112, 0.1013, 0.0529, 0.0593, 0.0542, and 0.0545; and PSNR improvements of 3.1967, 3.7965, 8.2017, 6.6921, 3.1426, 3.4365, 3.1193, and 3.1712, respectively.

Furthermore, Fig. 17 presents the interferograms generated between each result and the reference image. The interferograms for Adam-ADMM and SSI-net exhibit the clearest and most continuous fringes, confirming their exceptional ability to preserve phase information, thereby demonstrating their enhanced RFI suppression capability from another dimension. Besides, we calculate the coherence coefficient between the image processed by each method and the reference image. The coherence coefficient is defined as

$$\gamma = \frac{\sum_{i=1}^{Q_a}\sum_{j=1}^{Q_r}|\mu_1(i,j)|\,|\mu_2(i,j)|}{\sqrt{\sum_{i=1}^{Q_a}\sum_{j=1}^{Q_r}|\mu_1(i,j)|^2}\sqrt{\sum_{i=1}^{Q_a}\sum_{j=1}^{Q_r}|\mu_2(i,j)|^2}} \tag{29}$$

where μ_1 and μ_2 represent the two images. Q_a and Q_r denote the sampling numbers along azimuth and range dimensions, respectively.

The coherence coefficient analysis reveals two distinct trends. Compared to the original interferogram, the results processed by RPCA and FELS-net show a further degradation in coherence with the reference image. In contrast, all other methods achieve varying degrees of improvement. Specifically, Adam-ADMM improves the coherence coefficient by 0.1902, 0.1668, 0.1124, 0.1446, 0.1041, and 0.1898 compared to the FD, TFD, IMN-net, 2DSA, BSF, and Rand-ADMM methods, respectively. Similarly, SSI-net yields improvements of 0.2078, 0.1844, 0.13, 0.1622, 0.1609, and 0.2074 over the same set of baseline methods, respectively.

D. SAR Flight Experiment in Yantai

Our research group conducted another airborne SAR flight experiment in Yantai, Shandong, China, from September to October 2025. A Cessna-208 aircraft was used as the flight platform, carrying an SAR system independently developed by our group. During the experiment, the system transmitted an LFM signal with a bandwidth of 600 MHz and a pulsewidth of 8 μs. To prevent azimuth spectral aliasing of the echo, the PRF was set to 3200 Hz. The flight velocity of the aircraft platform was approximately 75.24 m/s. The detailed experimental parameters are shown in Table VIII.

The installation site of this experiment is shown in Fig. 18, where the various components can be seen, including the transmitter, rubidium clock, servo, servo power supply, power amplifier, inertial navigation system, and video monitoring subsystems.

The complex coastal electromagnetic environment unavoidably subjected the collected data to RFI. The nature of this RFI is analyzed in Fig. 19, which presents the time–frequency spectra of the received echoes at three distinct azimuth moments. Specifically, Fig. 19(a) and (b) illustrates contamination by LFM WBI, while Fig. 19(c) depicts a more complex scenario involving the simultaneous presence of both WBI and NBI. Due to its significantly higher energy relative to the target echoes, the RFI dominates and obscures useful targets in the image domain.

In the processing of the proposed framework, azimuth interception and range blocking are applied to the measured data to match the input dimensions of SSI-net. The SAR imaging result contaminated by RFI is presented in Fig. 20(a). To circumvent the noncooperative RFI, the second flight experiment was conducted in the early morning of the following day. Consequently, a high-quality, RFI-free SAR image was successfully acquired, as shown in Fig. 20(l). Fig. 20(b)–(k) shows the imaging results processed by different methods. Similarly, quantitative metrics, including NMSE, SSIM, and PSNR between the RFI-suppressed images and the reference image, are computed. The corresponding values are presented in Table IX and annotated in the top-left corner of each subfigure. Adam-ADMM and SSI-net deliver the best quantitative performance, as evidenced by their significantly lower NMSE and higher SSIM and PSNR values compared with other methods.

In addition, Fig. 21 presents the interferograms generated between each RFI suppressed images and the reference image. Among them, the results from Adam-ADMM and SSI-net exhibit the clearest and most continuous interference fringes, indicating superior phase preservation. Additionally, the coherence coefficients for three distinct island targets within

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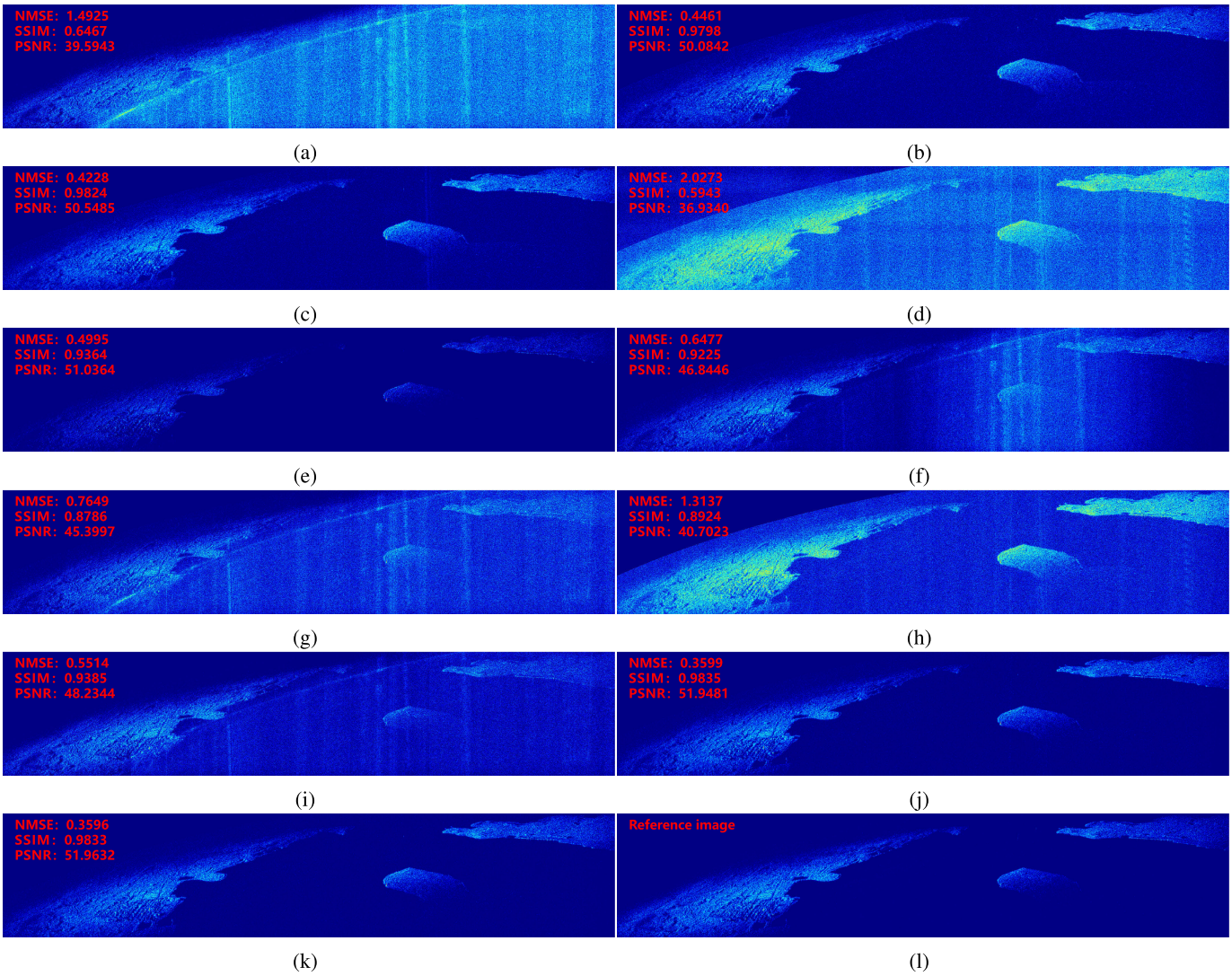

Fig. 20. Imaging results of flight experiment in Yantai. (a) Imaging result of the first day. (b) FD [21]. (c) TFD [22]. (d) RPCA [24]. (e) FELS-net [33]. (f) IMN-net [32]. (g) 2DSA [35]. (h) BSF [36]. (i) Rand-ADMM. (j) Adam-ADMM. (k) SSI-net. (l) Imaging result of the second day.

TABLE IX

PERFORMANCE OF DIFFERENT METHODS FOR YANTAI FLIGHT EXPERIMENT

Metrics	Before	FD	TFD	RPCA	FELS-net	IMN-net	2DSA	BSF	Rand-ADMM	Adam-ADMM	SSI-net
NMSE	1.4925	0.4461	0.4288	2.0273	0.4995	0.6477	0.7649	1.3137	0.5514	**0.3599**	**0.3596**
SSIM	0.6467	0.9798	0.9824	0.5943	0.9364	0.9225	0.8786	0.8924	0.9385	**0.9835**	**0.9833**
PSNR	39.6837	50.0842	50.5485	36.9340	51.0364	46.8446	45.3997	40.7023	48.2344	**51.9481**	**51.9632**

TABLE X

COHERENT COEFFICIENT FOR THE THREE ISLANDS IN YANTAI EXPERIMENT

Island label	Before	TD	TFD	RPCA	FELS-net	IMN-net	2DSA	BSF	Rand-ADMM	Adam-ADMM	SSI-net
Island 1	0.2600	0.3396	0.6348	0.5634	0.6085	0.8005	0.7521	0.8128	0.6158	**0.8129**	**0.8149**
Island 2	0.2021	0.4391	0.2127	0.2520	0.4892	0.2133	0.2380	0.3384	0.4443	**0.6385**	**0.6424**
Island 3	0.2336	0.3180	0.3591	0.3994	0.4023	0.5041	0.3564	0.4303	0.4054	**0.5303**	**0.5320**

each imaging result are evaluated quantitatively, as detailed in Table X. The three distinct island targets are labeled as island 1, island 2, and island 3 from left to right. It can be seen that, for island 1, the IMN-net, BSF, Adam-ADMM, and SSI-net methods all yield coherent coefficients above 0.8. For island 2, the coherent coefficients for IMN-net and

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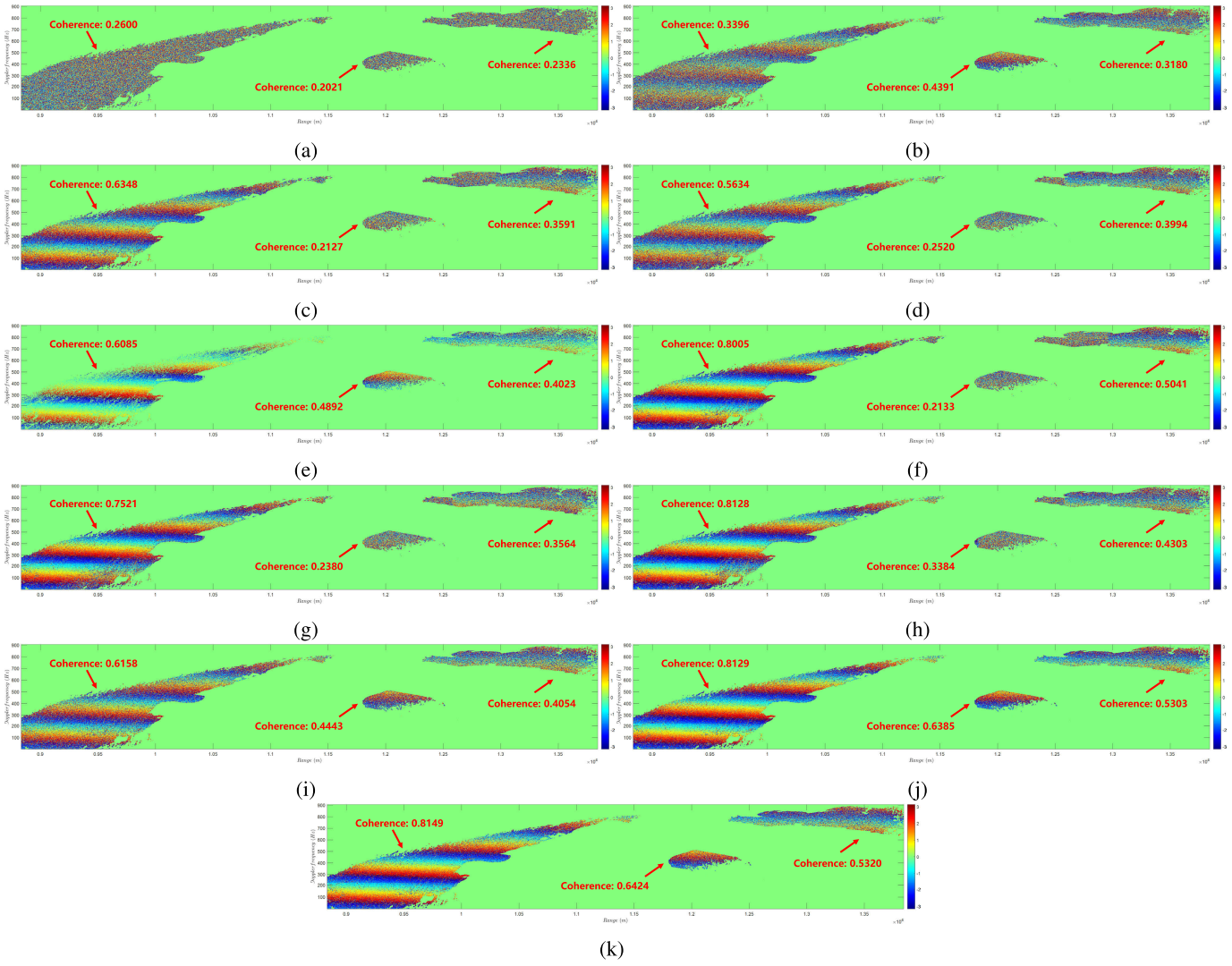

Fig. 21. Interferograms between the RFI suppressed results by different methods and the reference image in Yantai. (a) Before RFI suppression. (b) FD [21]. (c) TFD [22]. (d) RPCA [24]. (e) FELS-net [33]. (f) IMN-net [32]. (g) 2DSA [35]. (h) BSF [36]. (i) Rand-ADMM. (j) Adam-ADMM. (k) SSI-net.

BSF methods are below 0.35, in contrast to values exceeding 0.63 achieved by Adam-ADMM and SSI-net methods. For island 3, Adam-ADMM and SSI-net methods again performed best, with coherent coefficients greater than 0.53, surpassing all other methods. These results quantitatively prove that the proposed framework excels in preserving the target phase after interference suppression.

In summary, the proposed framework for simultaneous RFI separation and SAR imaging has been validated through comprehensive simulations and two airborne flight experiments, and comparative analyses against multiple baseline methods consistently demonstrate its superior performance in RFI suppression.

V. Conclusion

This article proposes a framework for simultaneous RFI separation and SAR imaging that requires no additional preprocessing or postprocessing. The core innovation lies in establishing a hybrid-domain-constrained model that bridges the echo domain and the image domain through the SAR imaging operator. By integrating this model with the ADMM iterative solution process, the specific SSI-net is constructed, which achieves effective separation of RFI and target echoes, thereby enabling the reconstruction of high-quality SAR images. Beyond that, for scenarios with limited or no training samples, a gradient-descent-based optimization algorithm is designed, achieving RFI suppression performance nearly on par with SSI-net at the cost of some computational efficiency. Furthermore, processing results from flight real measured data demonstrate that the proposed framework offers superior phase preservation capabilities, which are more conducive to extracting true target scattering characteristics.

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Jun'ao Li (Member, IEEE) received the B.S. degree in communication engineering from Zhengzhou University (ZZU), Zhengzhou, China, in 2019, and the Ph.D. degree from the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2025.

He is currently an Assistant Researcher with ZZU. His research interests include synthetic aperture radar (SAR), moving target indication (MTI), and localization.

Dr. Li was awarded the Zhongyuan Talent Program—Zhongyuan Outstanding Young Talents in 2026.

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Zhongyu Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2011 and 2017, respectively.

From 2015 to 2016, he was a Visiting Ph.D. Student with the Department of Information Engineering, Electronics and Telecommunications, Sapienza University of Rome, Rome, Italy. He is currently a Professor with UESTC. He has published more than 70 journal articles and conference papers. His research interests include synthetic aperture radar (SAR) technology (particular emphasis on bistatic/multistatic SAR moving target detection and imaging technologies).

Dr. Li was a recipient of the 2022 URSI Young Scientists Award, the 2022 Rising Star of Radar, and the Outstanding Doctoral Society Award of Chinese Institute of Electronics, and he was successfully selected for China Postdoctoral Innovative Talent Support Program. He received the Excellent Paper Award—First Prize at the 2019 6th APSAR. He served on the editorial boards of the *Modern Radar*, *Journal of Radars*, *Journal of Signal Processing*, *Telecommunication Engineering*, and *Radar Science and Technology*, and as a Guest Editor for *Remote Sensing* and *Applied Sciences*.

Jing Yang (Member, IEEE) received the B.S. degree in electronic information engineering and the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China, Chengdu, China, in 2017 and 2022, respectively.

From 2019 to 2020, she was a Visiting Researcher with the University of Naples Federico II, Naples, Italy. She is currently an Associate Professor with Zhengzhou University, Zhengzhou, China. Her research interests include cognitive radar waveform optimization and multifunction integrated radio frequency systems.

Dr. Yang received the Young Elite Scientists Sponsorship Program of Henan Province in 2025. She is currently a Youth Editorial Board Member of *Radar Science and Technology*.

Qing Yang (Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2018 and 2024, respectively.

He is currently an Assistant Researcher with UESTC. His research interests include synthetic aperture radar (SAR) imaging and moving target imaging (MTIm).

Junjie Wu (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2004, 2007, and 2013, respectively.

From January 2012 to January 2013, he was a Visiting Student with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. He is a Professor with UESTC. His research interests include synthetic aperture radar (SAR) imaging (particular emphasis on bistatic SAR).

Dr. Wu is a Reviewer of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRS), *IET Radar, Sonar & Navigation* (RSN), and so on.

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